MOEX Si thesis workflow — multi-scale CNN + ablations + selective tails + Grad-CAM + benchmarks

This Colab notebook is built for the thesis:

Intraday Decision Support for Traders: Explainable CNN-Based Directional Price Forecasting from Candlestick Chart Images

It is designed to cover the five main points discussed in the supervisor meeting:

1. **Multi-scale intraday input**
2. **Image-component ablations**
3. **Selective tail-rule decision support**
4. **Grad-CAM chart-vs-background analysis**
5. **Simple benchmark comparisons**

What this notebook does

- reads the **14 raw 1-minute MOEX Si contract candle CSVs** uploaded in Colab (left sidebar) or stored in Drive
- optionally reads the **FUTOI** file
- resamples to **15-minute OHLCV**
- selects the **front contract** by realized daily volume
- builds a clean continuous-front branch
- creates chronology-safe **master samples**
- trains:
  - **simple benchmark models**
  - **single-scale CNNs**
  - **image ablations**
  - **multi-scale CNN fusion**
- evaluates:
  - **full-sample metrics**
  - **tail-rule metrics**
  - **Grad-CAM chart-focused vs background-focused cases**

Important honesty note

This notebook is built to be **end-to-end workable** and to hit the requested research points. No notebook can honestly guarantee the **best possible score** in advance. It gives you a strong, reproducible framework to search for the best result on your data.

```
# Colab / Drive
from google.colab import drive
drive.mount('/content/drive')
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call dr
```

```
# ===== USER CONFIG =====
from pathlib import Path

# Change only if needed
PROJECT_ROOT = Path('/content/drive/MyDrive/moex_si_thesis_v2')
RAW_DIR = Path('/content')                      # uploaded files from left side
WORK_DIR = PROJECT_ROOT / 'work'
DERIVED_DIR = PROJECT_ROOT / 'derived'
RESULT_DIR = PROJECT_ROOT / 'results'
MODEL_DIR = PROJECT_ROOT / 'models'
FIG_DIR = PROJECT_ROOT / 'figures'
CACHE_DIR = PROJECT_ROOT / 'cache'

for p in [WORK_DIR, DERIVED_DIR, RESULT_DIR, MODEL_DIR, FIG_DIR, CACHE_DIR]:
    p.mkdir(parents=True, exist_ok=True)

# File patterns
SI_PATTERN = 'Si*_candles_2024-01-01_2025-12-31_1m.csv'
FUTOI_PATTERN = 'Si_futoi_2024-01-01_2025-12-31.csv'

# Thesis / experiment settings
BAR_FREQ = '15min'
HORIZON = 20
SCALES = [20, 60]           # multi-scale input; add 120 later if you want a he
MAX_LOOKBACK = max(SCALES)

# Intraday target rule
REQUIRE_FUTURE_SAME_DAY = True          # future target remains intraday
ALLOW_INPUT_CROSS_DAY = True            # needed for long contexts like I60

# q-filter
USE_Q_FILTER = True
Q_FILTER = 0.10                         # remove smallest 10% absolute train re

# Front selection / cleaning
MIN_DAILY_VOLUME = 100_000
```

```
EXCLUDE_ROLL_DAY = True

# Image settings
BAR_PIXELS = 3
HEIGHT_MAP = {20: 64, 60: 96, 120: 128}
PRICE_PANEL_RATIO = 0.80
VERTICAL_PADDING_FRAC = 0.03

# Training
SEED = 42
BATCH_SIZE = 64
MAX_EPOCHS = 20
PATIENCE = 5
NUM_WORKERS = 2
DEVICE_PREFERRED = 'cuda'

# Model search knobs
LR_SINGLE = 1e-4
LR_MULTI = 1e-4
WEIGHT_DECAY = 1e-4
THRESHOLD_METRIC = 'balanced_accuracy'   # threshold chosen on validation only
TAIL_GRID = [0.01, 0.015, 0.025, 0.05, 0.10]   # two-sided tails
```

```
# Imports
import os, gc, math, json, random, shutil, warnings
warnings.filterwarnings('ignore')

import numpy as np
import pandas as pd

from PIL import Image, ImageDraw
import matplotlib.pyplot as plt

from sklearn.metrics import (
    accuracy_score, balanced_accuracy_score, f1_score, roc_auc_score,
    matthews_corrcoef, precision_score, recall_score, confusion_matrix,
    log_loss, brier_score_loss
)
from sklearn.linear_model import LogisticRegression

import torch
import torch.nn as nn
from torch.utils.data import Dataset, DataLoader
from torchvision import transforms
from tqdm.auto import tqdm
```

```
# Reproducibility and device
def seed_everything(seed=42):
```

```
    random.seed(seed)
    np.random.seed(seed)
    torch.manual_seed(seed)
    torch.cuda.manual_seed_all(seed)

seed_everything(SEED)

DEVICE = torch.device('cuda' if (DEVICE_PREFERRED == 'cuda' and torch.cuda.is_a
print('Device:', DEVICE)
if DEVICE.type == 'cuda':
    print('GPU:', torch.cuda.get_device_name(0))
```

```
Device: cuda
GPU: Tesla T4
```

1) Discover and optionally copy the uploaded raw files

Upload the **14 contract CSVs** and the **FUTOI** file in the left Colab sidebar first. Then run the next cells.

```
# Discover raw files
si_files = sorted(RAW_DIR.glob(SI_PATTERN))
futoi_files = sorted(RAW_DIR.glob(FUTOI_PATTERN))

print('Found SI files:', len(si_files))
for f in si_files[:5]:
    print('  ', f.name)
print('Found FUTOI files:', len(futoi_files), [f.name for f in futoi_files])

assert len(si_files) >= 8, 'Upload the raw SI candle files first.'
```

```
Found SI files: 14
   SiH4_candles_2024-01-01_2025-12-31_1m.csv
   SiH5_candles_2024-01-01_2025-12-31_1m.csv
   SiH6_candles_2024-01-01_2025-12-31_1m.csv
   SiH7_candles_2024-01-01_2025-12-31_1m.csv
   SiM4_candles_2024-01-01_2025-12-31_1m.csv
Found FUTOI files: 1 ['Si_futoi_2024-01-01_2025-12-31.csv']
```

```
# Optional backup to Drive
BACKUP_RAW_TO_DRIVE = False

if BACKUP_RAW_TO_DRIVE:
    raw_backup = PROJECT_ROOT / 'raw_backup'
    raw_backup.mkdir(parents=True, exist_ok=True)
    for fp in si_files + futoi_files:
        dst = raw_backup / fp.name
```

```
        if not dst.exists():
            shutil.copy2(fp, dst)
    print('Raw files copied to', raw_backup)
else:
    print('Skipping raw backup copy.')
```

```
Skipping raw backup copy.
```

2) Load and standardize the raw candle files

This section merges the raw 1-minute contract files into one long table and then resamples to 15-minute bars.

```
def standardize_candle_csv(fp: Path) -> pd.DataFrame:
    df = pd.read_csv(fp)
    cols = {c.lower(): c for c in df.columns}

    # Required columns in your MOEX files are expected to be present under thes
    rename_map = {}
    for want in ['begin', 'open', 'high', 'low', 'close', 'volume', 'value', 's
        for c in df.columns:
            if c.lower() == want:
                rename_map[c] = want
                break
    df = df.rename(columns=rename_map)

    required = ['begin', 'open', 'high', 'low', 'close', 'volume']
    missing = [c for c in required if c not in df.columns]
    if missing:
        raise ValueError(f'{fp.name}: missing columns {missing}')

    if 'secid' not in df.columns:
        # infer from file name
        secid = fp.name.split('_candles_')[0]
        df['secid'] = secid

    if 'value' not in df.columns:
        df['value'] = 0.0

    df['begin'] = pd.to_datetime(df['begin'])
    df = df.sort_values('begin').reset_index(drop=True)

    keep = ['begin', 'open', 'high', 'low', 'close', 'volume', 'value', 'secid'
    return df[keep].copy()

raw_fp = DERIVED_DIR / 'si_1m_merged_raw.parquet'
if raw_fp.exists():
```

```
    raw_1m = pd.read_parquet(raw_fp)
else:
    parts = []
    for fp in tqdm(si_files, desc='Loading raw candle files'):
        x = standardize_candle_csv(fp)
        x['contract'] = x['secid']
        parts.append(x)
    raw_1m = pd.concat(parts, ignore_index=True)
    raw_1m['trade_date'] = raw_1m['begin'].dt.date
    raw_1m.to_parquet(raw_fp, index=False)

raw_1m.head(), raw_1m.shape
```

```
Loading raw candle files: 100%                                   14/14 [00:01<00:00,  7.18it/s]
(                begin   open   high    low  close  volume  value secid  \
 0 2024-01-03 08:59:00  92000  92000  91965  92000    1982      0  SiH4
 1 2024-01-03 09:00:00  92002  92100  92002  92073    4407      0  SiH4
 2 2024-01-03 09:01:00  92072  92088  92032  92084    1409      0  SiH4
 3 2024-01-03 09:02:00  92079  92097  92076  92096     560      0  SiH4
 4 2024-01-03 09:03:00  92095  92110  92085  92095     946      0  SiH4

   contract  trade_date
 0     SiH4  2024-01-03
 1     SiH4  2024-01-03
 2     SiH4  2024-01-03
 3     SiH4  2024-01-03
 4     SiH4  2024-01-03  ,
 (775621, 10))
```

```
def resample_contract_to_15m(df_contract: pd.DataFrame, freq='15min') -> pd.Dat
    x = df_contract.sort_values('begin').copy()
    x = x.set_index('begin')

    out = x.resample(freq, label='left', closed='left').agg({
        'open': 'first',
        'high': 'max',
        'low': 'min',
        'close': 'last',
        'volume': 'sum',
        'value': 'sum',
        'secid': 'last',
        'contract': 'last',
    }).dropna(subset=['open', 'high', 'low', 'close']).reset_index()

    out['trade_date'] = out['begin'].dt.date
    return out

resampled_fp = DERIVED_DIR / 'si_candles_15m_by_contract.parquet'
if resampled_fp.exists():
    si_15m = pd.read_parquet(resampled_fp)
```

```
else:
    chunks = []
    for secid, g in tqdm(raw_1m.groupby('contract'), desc='Resampling by contra
        chunks.append(resample_contract_to_15m(g, BAR_FREQ))
    si_15m = pd.concat(chunks, ignore_index=True)
    si_15m.to_parquet(resampled_fp, index=False)

si_15m.head(), si_15m.shape
```

```
Resampling by contract: 100%                                    14/14 [00:00<00:00, 26.68it/s]
(                begin     open     high      low    close  volume  value  \
 0 2024-01-03 08:45:00  92000.0  92000.0  91965.0  92000.0    1982      0
 1 2024-01-03 09:00:00  92002.0  92185.0  92002.0  92181.0   14431      0
 2 2024-01-03 09:15:00  92183.0  92218.0  92123.0  92173.0    5678      0
 3 2024-01-03 09:30:00  92172.0  92207.0  92155.0  92156.0    2936      0
 4 2024-01-03 09:45:00  92156.0  92170.0  92142.0  92170.0    1884      0

   secid contract  trade_date
 0  SiH4     SiH4  2024-01-03
 1  SiH4     SiH4  2024-01-03
 2  SiH4     SiH4  2024-01-03
 3  SiH4     SiH4  2024-01-03
 4  SiH4     SiH4  2024-01-03  ,
 (89628, 10))
```

3) Front-contract selection and clean continuous-front branch

The front contract is chosen by **highest realized daily volume**. Then a liquidity filter removes weak days.

```
# Daily realized volume per contract
front_daily_fp = DERIVED_DIR / 'si_front_daily_contracts_clean.parquet'
cf_fp = DERIVED_DIR / 'si_continuous_front_15m_clean_v2.parquet'

if front_daily_fp.exists() and cf_fp.exists():
    front_daily = pd.read_parquet(front_daily_fp)
    cf = pd.read_parquet(cf_fp)
else:
    daily_vol = (
        si_15m.groupby(['trade_date', 'contract'], as_index=False)
             .agg(daily_volume=('volume', 'sum'))
    )

    # Pick front by max daily volume
    front_daily = (
        daily_vol.sort_values(['trade_date', 'daily_volume', 'contract'], ascer
```

```
                    .drop_duplicates('trade_date')
                    .rename(columns={'contract': 'front_contract'})
    )

    # Liquidity filter
    front_daily = front_daily[front_daily['daily_volume'] >= MIN_DAILY_VOLUME].
    front_daily = front_daily.sort_values('trade_date').reset_index(drop=True)
    front_daily['prev_front'] = front_daily['front_contract'].shift(1)
    front_daily['is_roll_day'] = front_daily['front_contract'] != front_daily['
    front_daily.loc[front_daily.index[0], 'is_roll_day'] = True

    # Join back
    cf = si_15m.merge(front_daily[['trade_date', 'front_contract', 'is_roll_day
    cf = cf[cf['contract'] == cf['front_contract']].copy()
    cf = cf.sort_values('begin').reset_index(drop=True)

    front_daily.to_parquet(front_daily_fp, index=False)
    cf.to_parquet(cf_fp, index=False)

front_daily.head(), cf.shape
```

```
(   trade_date front_contract  daily_volume prev_front  is_roll_day
 0  2024-01-03           SiH4        438437       None         True
 1  2024-01-04           SiH4        508126       SiH4        False
 2  2024-01-05           SiH4        489735       SiH4        False
 3  2024-01-08           SiH4        726473       SiH4        False
 4  2024-01-09           SiH4        943599       SiH4        False,
 (29926, 12))
```

```
# Quick diagnostics
print('Front contracts used:', sorted(cf['contract'].unique()))
print('Trade dates:', cf['trade_date'].min(), 'to', cf['trade_date'].max(), '|
print('Rows:', len(cf))
print('Rows by contract:')
display(cf['contract'].value_counts().sort_index().to_frame('rows'))
```

```
Front contracts used: ['SiH4', 'SiH5', 'SiM4', 'SiM5', 'SiU4', 'SiU5', 'SiZ4', '
Trade dates: 2024-01-03 to 2025-12-18 | n = 502
Rows: 29926
Rows by contract:
```

contract	rows
SiH4	3294
SiH5	3686
SiM4	3823
SiM5	3776
SiU4	3685
SiU5	3898
SiZ4	3702
SiZ5	4062

4) Build a master sample index

We build one **master manifest** using the largest lookback. Any smaller scale is later derived from the same anchor timestamp, so all scales stay aligned for fusion.

```
def build_master_manifest(
    cf: pd.DataFrame,
    max_lookback: int,
    horizon: int,
    require_future_same_day: bool = True,
    allow_input_cross_day: bool = True,
    exclude_roll_day: bool = True,
):
    x = cf.copy().reset_index(drop=True)
    rows = []
    n = len(x)

    for anchor_idx in tqdm(range(max_lookback - 1, n - horizon), desc='Building
        anchor = x.iloc[anchor_idx]
        future = x.iloc[anchor_idx + horizon]

        start_idx = anchor_idx - max_lookback + 1
        end_idx = anchor_idx + horizon

        block = x.iloc[start_idx:end_idx + 1]
```

```
            # Full path must stay in one contract
            if block['contract'].nunique() != 1:
                continue

            # Optionally keep intraday target
            if require_future_same_day and future['trade_date'] != anchor['trade_da
                continue

            # Optionally force input to stay same day too
            if not allow_input_cross_day:
                if x.iloc[start_idx:anchor_idx + 1]['trade_date'].nunique() != 1:
                    continue

            # Avoid roll contamination
            if exclude_roll_day and block['is_roll_day'].any():
                continue

            ret = (future['close'] / anchor['close']) - 1.0
            rows.append({
                'sample_id': len(rows),
                'anchor_idx': int(anchor_idx),
                'future_idx': int(anchor_idx + horizon),
                'anchor_ts': anchor['begin'],
                'future_ts': future['begin'],
                'trade_date': anchor['trade_date'],
                'future_trade_date': future['trade_date'],
                'contract': anchor['contract'],
                'forward_return': float(ret),
                'label_up': int(ret > 0),
            })

        return pd.DataFrame(rows)

    master_fp = DERIVED_DIR / f'master_manifest_I{MAX_LOOKBACK}_R{HORIZON}.parquet'
    if master_fp.exists():
        master = pd.read_parquet(master_fp)
    else:
        master = build_master_manifest(
            cf=cf,
            max_lookback=MAX_LOOKBACK,
            horizon=HORIZON,
            require_future_same_day=REQUIRE_FUTURE_SAME_DAY,
            allow_input_cross_day=ALLOW_INPUT_CROSS_DAY,
            exclude_roll_day=EXCLUDE_ROLL_DAY,
        )
        master.to_parquet(master_fp, index=False)

    master.head(), master.shape
```

```
Building master manifest: 100%                          29847/29847 [00:12<00:00, 2796.03it/s]
(   sample_id  anchor_idx  future_idx           anchor_ts           future_ts
\
 0          0         122         142 2024-01-05 08:45:00 2024-01-05 13:45:00
 1          1         123         143 2024-01-05 09:00:00 2024-01-05 14:00:00
 2          2         124         144 2024-01-05 09:15:00 2024-01-05 14:15:00
 3          3         125         145 2024-01-05 09:30:00 2024-01-05 14:30:00
 4          4         126         146 2024-01-05 09:45:00 2024-01-05 14:45:00

    trade_date future_trade_date contract  forward_return  label_up
 0  2024-01-05        2024-01-05     SiH4       -0.006706         0
 1  2024-01-05        2024-01-05     SiH4       -0.008950         0
 2  2024-01-05        2024-01-05     SiH4       -0.007553         0
 3  2024-01-05        2024-01-05     SiH4       -0.009006         0
 4  2024-01-05        2024-01-05     SiH4       -0.009470         0  ,
 (19249, 10))
```

```
# Strict chronological split by unique trade date
dates = sorted(pd.Series(master['trade_date']).unique())
n_dates = len(dates)
train_end = int(0.70 * n_dates)
val_end = int(0.85 * n_dates)

train_dates = set(dates[:train_end])
val_dates = set(dates[train_end:val_end])
test_dates = set(dates[val_end:])

def assign_split(d):
    if d in train_dates:
        return 'train'
    if d in val_dates:
        return 'val'
    return 'test'

master['split'] = master['trade_date'].map(assign_split)

# q-filter based on training only
if USE_Q_FILTER:
    q_thr = master.loc[master['split'] == 'train', 'forward_return'].abs().quar
    master['keep_q'] = master['forward_return'].abs() > q_thr
else:
    q_thr = 0.0
    master['keep_q'] = True

master_q = master[master['keep_q']].copy().reset_index(drop=True)
print('Q-filter threshold:', q_thr)
print(master_q['split'].value_counts())
display(master_q.groupby('split')['label_up'].mean().to_frame('positive_rate'))
```

```
Q-filter threshold: 0.00038779644882541714
split
train    11964
test      2722
val       2617
Name: count, dtype: int64
```

split	positive_rate
test	0.421014
train	0.481862
val	0.436760

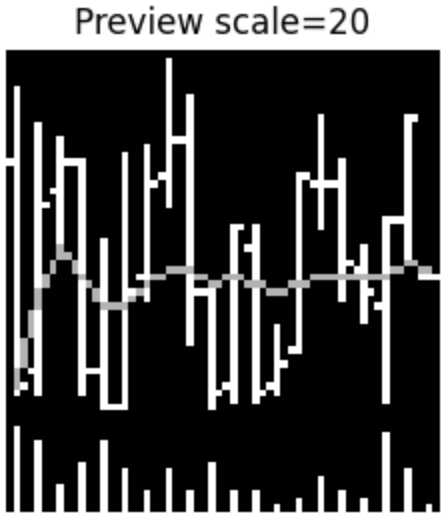
5) Render reference-style grayscale chart images

We render **OHLC bars**, optional **moving average**, and optional **volume panel**. This lets us test the image-component ablations cleanly.

```
def get_shape_for_scale(scale: int):
    h = HEIGHT_MAP.get(scale, 96)
    w = BAR_PIXELS * scale
    return h, w

def render_refstyle_window(
    window_df: pd.DataFrame,
    scale: int,
    include_ma: bool = True,
    include_volume: bool = True,
    bg_color=0,
    fg_color=255,
):
    H, W = get_shape_for_scale(scale)
    price_h = int(round(H * PRICE_PANEL_RATIO))
    vol_h = H - price_h

    img = Image.new('L', (W, H), color=bg_color)
    draw = ImageDraw.Draw(img)

    wdf = window_df.tail(scale).copy().reset_index(drop=True)

    # Price scaling
    px_min = wdf['low'].min()
    px_max = wdf['high'].max()
    px_rng = max(float(px_max - px_min), 1e-9)
    pad = px_rng * VERTICAL_PADDING_FRAC
```

```
        px_low = px_min - pad
        px_high = px_max + pad
        px_den = max(px_high - px_low, 1e-9)

        def y_price(v):
            # invert y
            return int(round((price_h - 1) - (float(v) - px_low) / px_den * (price_

        # Volume scaling
        vmax = max(float(wdf['volume'].max()), 1.0)

        # Draw OHLC bars
        for i, row in wdf.iterrows():
            x0 = i * BAR_PIXELS
            xc = x0 + 1
            xl = x0
            xr = x0 + 2

            yh = y_price(row['high'])
            yl = y_price(row['low'])
            yo = y_price(row['open'])
            yc = y_price(row['close'])

            draw.line([(xc, yh), (xc, yl)], fill=fg_color, width=1)
            draw.point((xl, yo), fill=fg_color)
            draw.point((xr, yc), fill=fg_color)

            if include_volume and vol_h > 0:
                vh = int(round((float(row['volume']) / vmax) * max(vol_h - 1, 1)))
                v_top = H - vh
                draw.line([(xc, H - 1), (xc, v_top)], fill=fg_color, width=1)

        # Draw moving average over close
        if include_ma:
            ma = wdf['close'].rolling(scale, min_periods=1).mean()
            pts = []
            for i, v in enumerate(ma):
                x = i * BAR_PIXELS + 1
                y = y_price(v)
                pts.append((x, y))
            if len(pts) >= 2:
                draw.line(pts, fill=180, width=1)  # gray line

        return img

    # Example preview
    ex = master_q.iloc[0]
    anchor_idx = int(ex['anchor_idx'])
    scale = SCALES[0]
    window = cf.iloc[anchor_idx - scale + 1: anchor_idx + 1].copy()
    preview = render_refstyle_window(window, scale=scale, include_ma=True, include_
```

```
plt.figure(figsize=(8, 3))
plt.imshow(preview, cmap='gray')
plt.axis('off')
plt.title(f'Preview scale={scale}')
plt.show()
```




```
# Optional image cache for speed
def get_cache_subdir(scale, include_ma, include_volume):
    tag = f'I{scale}_ma{int(include_ma)}_vol{int(include_volume)}'
    out = CACHE_DIR / tag
    out.mkdir(parents=True, exist_ok=True)
    return out

def pre_render_cache(master_df, scales, ablation_grid):
    for scale in scales:
        for include_ma, include_volume in ablation_grid:
            out_dir = get_cache_subdir(scale, include_ma, include_volume)
            for _, r in tqdm(master_df.iterrows(), total=len(master_df), desc=f
                fp = out_dir / f"{int(r['sample_id'])}.png"
                if fp.exists():
                    continue
                a = int(r['anchor_idx'])
                window = cf.iloc[a - scale + 1: a + 1].copy()
                img = render_refstyle_window(window, scale=scale, include_ma=ir
                img.save(fp)

ABLATION_GRID = [(False, False), (True, False), (False, True), (True, True)]
PRE_RENDER_IMAGES = False   # set True for faster repeated runs at the cost of

if PRE_RENDER_IMAGES:
    pre_render_cache(master_q, SCALES, ABLATION_GRID)
else:
    print('Skipping pre-render cache.')
```

```
Skipping pre-render cache.
```

6) Datasets and models

We define:

- a **single-scale CNN**
- a **multi-scale fusion CNN**
- a **simple numeric benchmark feature set**

```
# Torch transforms
to_tensor = transforms.Compose([
    transforms.ToTensor(),   # [0,1], shape [1,H,W]
])

class SingleScaleChartDataset(Dataset):
    def __init__(self, manifest_df, cf_df, scale, include_ma=True, include_volu
        self.df = manifest_df.reset_index(drop=True)
        self.cf = cf_df
        self.scale = scale
        self.include_ma = include_ma
        self.include_volume = include_volume
        self.use_cache = use_cache
        self.cache_dir = get_cache_subdir(scale, include_ma, include_volume)

    def __len__(self):
        return len(self.df)

    def __getitem__(self, idx):
        r = self.df.iloc[idx]
        sid = int(r['sample_id'])
        if self.use_cache and (self.cache_dir / f'{sid}.png').exists():
            img = Image.open(self.cache_dir / f'{sid}.png').convert('L')
        else:
            a = int(r['anchor_idx'])
            window = self.cf.iloc[a - self.scale + 1: a + 1].copy()
            img = render_refstyle_window(window, self.scale, self.include_ma, s
        x = to_tensor(img)
        y = int(r['label_up'])
        return {
            'x': x,
            'y': y,
            'sample_id': sid,
            'anchor_ts': r['anchor_ts'],
            'contract': r['contract'],
            'forward_return': float(r['forward_return']),
        }
```

```
class MultiScaleChartDataset(Dataset):
    def __init__(self, manifest_df, cf_df, scales, include_ma=True, include_vol
        self.df = manifest_df.reset_index(drop=True)
        self.cf = cf_df
        self.scales = list(scales)
        self.include_ma = include_ma
        self.include_volume = include_volume
        self.use_cache = use_cache
        self.cache_dirs = {s: get_cache_subdir(s, include_ma, include_volume) f

    def __len__(self):
        return len(self.df)

    def __getitem__(self, idx):
        r = self.df.iloc[idx]
        sid = int(r['sample_id'])
        out = {}
        for s in self.scales:
            fp = self.cache_dirs[s] / f'{sid}.png'
            if self.use_cache and fp.exists():
                img = Image.open(fp).convert('L')
            else:
                a = int(r['anchor_idx'])
                window = self.cf.iloc[a - s + 1: a + 1].copy()
                img = render_refstyle_window(window, s, self.include_ma, self.i
            out[f'x_{s}'] = to_tensor(img)
        out['y'] = int(r['label_up'])
        out['sample_id'] = sid
        out['anchor_ts'] = r['anchor_ts']
        out['contract'] = r['contract']
        out['forward_return'] = float(r['forward_return'])
        return out
```

```
class CNNFeatureExtractor(nn.Module):
    def __init__(self, in_channels=1, base=32, depth=4):
        super().__init__()
        channels = []
        c = base
        for i in range(depth):
            channels.append(c)
            c = min(c * 2, 256)

        blocks = []
        prev = in_channels
        for c in channels:
            blocks.extend([
                nn.Conv2d(prev, c, kernel_size=3, padding=1, bias=False),
                nn.BatchNorm2d(c),
```

```
                nn.LeakyReLU(0.01, inplace=True),
                nn.MaxPool2d(2, 2),
            ])
            prev = c
        self.backbone = nn.Sequential(*blocks)
        self.pool = nn.AdaptiveAvgPool2d((4, 4))
        self.out_dim = prev * 4 * 4

    def forward(self, x):
        z = self.backbone(x)
        z = self.pool(z)
        z = torch.flatten(z, 1)
        return z

def depth_for_scale(scale):
    if scale <= 20:
        return 3
    elif scale <= 60:
        return 4
    else:
        return 5

class RefStyleCNN(nn.Module):
    def __init__(self, scale, dropout=0.5):
        super().__init__()
        self.scale = scale
        self.fe = CNNFeatureExtractor(depth=depth_for_scale(scale))
        self.head = nn.Sequential(
            nn.Dropout(dropout),
            nn.Linear(self.fe.out_dim, 2),
        )
    def forward(self, x):
        z = self.fe(x)
        return self.head(z)

class MultiScaleFusionCNN(nn.Module):
    def __init__(self, scales, dropout=0.5):
        super().__init__()
        self.scales = list(scales)
        self.branches = nn.ModuleDict({
            str(s): CNNFeatureExtractor(depth=depth_for_scale(s)) for s in scal
        })
        total_dim = sum(self.branches[str(s)].out_dim for s in scales)
        self.head = nn.Sequential(
            nn.Dropout(dropout),
            nn.Linear(total_dim, 128),
            nn.LeakyReLU(0.01, inplace=True),
            nn.Dropout(dropout),
            nn.Linear(128, 2),
        )
```

```
    def forward(self, batch_x_dict):
        feats = []
        for s in self.scales:
            feats.append(self.branches[str(s)](batch_x_dict[f'x_{s}']))
        z = torch.cat(feats, dim=1)
        return self.head(z)
```

```
# Numeric benchmark features
def build_numeric_features(master_df, cf_df):
    rows = []
    for _, r in tqdm(master_df.iterrows(), total=len(master_df), desc='Building
        a = int(r['anchor_idx'])
        x20 = cf_df.iloc[a - 20 + 1: a + 1].copy()
        x60 = cf_df.iloc[a - 60 + 1: a + 1].copy()

        close20 = x20['close'].values
        close60 = x60['close'].values

        def safe_ret(arr):
            return float(arr[-1] / arr[0] - 1.0) if len(arr) > 1 and arr[0] !=

        feat = {
            'sample_id': int(r['sample_id']),
            'label_up': int(r['label_up']),
            'split': r['split'],
            'forward_return': float(r['forward_return']),
            'contract': r['contract'],
            'mom_5h': safe_ret(close20),
            'mom_15h': safe_ret(close60),
            'range_20': float((x20['high'].max() - x20['low'].min()) / max(abs(
            'close_vs_ma20': float((x20['close'].iloc[-1] / max(x20['close'].rc
            'vol_ratio_20': float(x20['volume'].iloc[-1] / max(x20['volume'].rc
            'last_bar_body': float((x20['close'].iloc[-1] - x20['open'].iloc[-1
            'last_bar_range': float((x20['high'].iloc[-1] - x20['low'].iloc[-1]
        }
        rows.append(feat)
    return pd.DataFrame(rows)

bench_fp = DERIVED_DIR / 'numeric_bench_features.parquet'
if bench_fp.exists():
    bench_df = pd.read_parquet(bench_fp)
else:
    bench_df = build_numeric_features(master_q, cf)
    bench_df.to_parquet(bench_fp, index=False)

bench_df.head()
```

```
Building numeric features: 100%                          17303/17303 [00:16<00:00, 1101.63it/s]
```

	sample_id	label_up	split	forward_return	contract	mom_5h	mom_15h	rang
0	0	0	train	-0.006706	SiH4	0.000282	-0.009746	0.0(
1	1	0	train	-0.008950	SiH4	0.001172	-0.006983	0.0(
2	2	0	train	-0.007553	SiH4	0.001063	-0.006973	0.0(
3	3	0	train	-0.009006	SiH4	0.002670	-0.003065	0.0(
4	4	0	train	-0.009470	SiH4	0.002898	-0.003645	0.0(

7) Training / evaluation helpers

These helpers train the models, choose the best checkpoint by validation AUC, tune the full-sample threshold on validation only, and then evaluate on test.

```
def metric_bundle(y_true, y_prob, threshold=0.5):
    y_true = np.asarray(y_true).astype(int)
    y_prob = np.asarray(y_prob).astype(float)
    y_pred = (y_prob >= threshold).astype(int)

    out = {
        'rows': int(len(y_true)),
        'threshold': float(threshold),
        'coverage': 1.0,
        'pred_positive_rate': float(y_pred.mean()) if len(y_pred) else np.nan,
        'true_positive_rate': float(y_true.mean()) if len(y_true) else np.nan,
        'accuracy': accuracy_score(y_true, y_pred),
        'balanced_accuracy': balanced_accuracy_score(y_true, y_pred),
        'precision': precision_score(y_true, y_pred, zero_division=0),
        'recall': recall_score(y_true, y_pred, zero_division=0),
        'f1': f1_score(y_true, y_pred, zero_division=0),
        'mcc': matthews_corrcoef(y_true, y_pred) if len(np.unique(y_pred)) > 1
        'roc_auc': roc_auc_score(y_true, y_prob) if len(np.unique(y_true)) > 1
        'log_loss': log_loss(y_true, np.clip(y_prob, 1e-6, 1-1e-6), labels=[0,1
        'brier': brier_score_loss(y_true, y_prob),
    }
    tn, fp, fn, tp = confusion_matrix(y_true, y_pred).ravel()
    out.update({'tn': int(tn), 'fp': int(fp), 'fn': int(fn), 'tp': int(tp)})
    return out

def choose_threshold(y_true, y_prob, metric='balanced_accuracy'):
    grid = np.linspace(0.05, 0.95, 181)
    best_t, best_score = 0.5, -np.inf
    for t in grid:
```

```
            m = metric_bundle(y_true, y_prob, threshold=t)
            score = m[metric]
            if np.isnan(score):
                continue
            if score > best_score:
                best_score = score
                best_t = float(t)
        return best_t, best_score

    def evaluate_tail_rule(y_true, y_prob, top_q=0.025, bottom_q=0.025):
        y_true = np.asarray(y_true).astype(int)
        y_prob = np.asarray(y_prob).astype(float)
        low_thr = np.quantile(y_prob, bottom_q)
        high_thr = np.quantile(y_prob, 1 - top_q)

        keep_short = y_prob <= low_thr
        keep_long = y_prob >= high_thr
        keep = keep_short | keep_long

        if keep.sum() == 0:
            return {'coverage': 0.0}

        y_pred_tail = np.where(keep_long[keep], 1, 0)
        y_true_tail = y_true[keep]
        out = {
            'coverage': float(keep.mean()),
            'rows_kept': int(keep.sum()),
            'tail_low_thr': float(low_thr),
            'tail_high_thr': float(high_thr),
            'accuracy': accuracy_score(y_true_tail, y_pred_tail),
            'balanced_accuracy': balanced_accuracy_score(y_true_tail, y_pred_tail),
            'precision': precision_score(y_true_tail, y_pred_tail, zero_division=0)
            'recall': recall_score(y_true_tail, y_pred_tail, zero_division=0),
            'f1': f1_score(y_true_tail, y_pred_tail, zero_division=0),
            'mcc': matthews_corrcoef(y_true_tail, y_pred_tail) if len(np.unique(y_p
        }
        return out
```

```
    def run_epoch_single(model, loader, optimizer=None, class_weights=None):
        train_mode = optimizer is not None
        model.train(train_mode)
        criterion = nn.CrossEntropyLoss(weight=class_weights)

        losses = []
        probs = []
        ys = []
        ids = []

        for batch in tqdm(loader, leave=False):
```

```
            x = batch['x'].to(DEVICE, non_blocking=True)
            y = batch['y'].to(DEVICE, non_blocking=True)

            with torch.set_grad_enabled(train_mode):
                logits = model(x)
                loss = criterion(logits, y)
                if train_mode:
                    optimizer.zero_grad(set_to_none=True)
                    loss.backward()
                    optimizer.step()

            p = torch.softmax(logits, dim=1)[:, 1].detach().cpu().numpy()
            losses.append(float(loss.item()))
            probs.extend(p.tolist())
            ys.extend(batch['y'].numpy().tolist())
            ids.extend(batch['sample_id'].numpy().tolist())

        return {
            'loss': float(np.mean(losses)),
            'y_true': np.array(ys),
            'y_prob': np.array(probs),
            'sample_id': np.array(ids),
        }

    def run_epoch_multi(model, loader, optimizer=None, class_weights=None):
        train_mode = optimizer is not None
        model.train(train_mode)
        criterion = nn.CrossEntropyLoss(weight=class_weights)

        losses = []
        probs = []
        ys = []
        ids = []

        for batch in tqdm(loader, leave=False):
            xdict = {k: v.to(DEVICE, non_blocking=True) for k, v in batch.items() i
            y = batch['y'].to(DEVICE, non_blocking=True)

            with torch.set_grad_enabled(train_mode):
                logits = model(xdict)
                loss = criterion(logits, y)
                if train_mode:
                    optimizer.zero_grad(set_to_none=True)
                    loss.backward()
                    optimizer.step()

            p = torch.softmax(logits, dim=1)[:, 1].detach().cpu().numpy()
            losses.append(float(loss.item()))
            probs.extend(p.tolist())
            ys.extend(batch['y'].numpy().tolist())
            ids.extend(batch['sample_id'].numpy().tolist())
```

```
    return {
        'loss': float(np.mean(losses)),
        'y_true': np.array(ys),
        'y_prob': np.array(probs),
        'sample_id': np.array(ids),
    }
```

```
def train_single_scale(
    manifest_df,
    scale,
    include_ma=True,
    include_volume=True,
    lr=1e-4,
    weight_decay=1e-4,
    max_epochs=20,
    patience=5,
    use_cache=False,
    run_name='single',
):
    train_df = manifest_df[manifest_df['split'] == 'train'].copy()
    val_df = manifest_df[manifest_df['split'] == 'val'].copy()
    test_df = manifest_df[manifest_df['split'] == 'test'].copy()

    ds_train = SingleScaleChartDataset(train_df, cf, scale, include_ma, include
    ds_val = SingleScaleChartDataset(val_df, cf, scale, include_ma, include_vol
    ds_test = SingleScaleChartDataset(test_df, cf, scale, include_ma, include_v

    dl_train = DataLoader(ds_train, batch_size=BATCH_SIZE, shuffle=True, num_wo
    dl_val = DataLoader(ds_val, batch_size=BATCH_SIZE, shuffle=False, num_worke
    dl_test = DataLoader(ds_test, batch_size=BATCH_SIZE, shuffle=False, num_wor

    model = RefStyleCNN(scale).to(DEVICE)
    optimizer = torch.optim.AdamW(model.parameters(), lr=lr, weight_decay=weigh

    pos = train_df['label_up'].mean()
    w0 = 1.0 / max(1.0 - pos, 1e-6)
    w1 = 1.0 / max(pos, 1e-6)
    class_weights = torch.tensor([w0, w1], dtype=torch.float32, device=DEVICE)

    best_auc = -np.inf
    best_state = None
    best_epoch = -1
    bad = 0
    history = []

    for epoch in range(1, max_epochs + 1):
        tr = run_epoch_single(model, dl_train, optimizer=optimizer, class_weigh
        va = run_epoch_single(model, dl_val, optimizer=None, class_weights=clas
```

```
        tr_m = metric_bundle(tr['y_true'], tr['y_prob'], threshold=0.5)
        va_m = metric_bundle(va['y_true'], va['y_prob'], threshold=0.5)
        history.append({
            'epoch': epoch,
            'train_loss': tr['loss'],
            'val_loss': va['loss'],
            'train_auc': tr_m['roc_auc'],
            'val_auc': va_m['roc_auc'],
            'train_acc': tr_m['accuracy'],
            'val_acc': va_m['accuracy'],
        })

        if va_m['roc_auc'] > best_auc:
            best_auc = va_m['roc_auc']
            best_state = {k: v.detach().cpu() for k, v in model.state_dict().it
            best_epoch = epoch
            bad = 0
        else:
            bad += 1
            if bad >= patience:
                break

    model.load_state_dict(best_state)
    va = run_epoch_single(model, dl_val, optimizer=None, class_weights=class_we
    te = run_epoch_single(model, dl_test, optimizer=None, class_weights=class_w

    best_thr, _ = choose_threshold(va['y_true'], va['y_prob'], metric=THRESHOLD
    val_metrics = metric_bundle(va['y_true'], va['y_prob'], threshold=best_thr)
    test_metrics = metric_bundle(te['y_true'], te['y_prob'], threshold=best_thr

    out_pred = test_df[['sample_id', 'anchor_ts', 'contract', 'forward_return',
    out_pred['prob_up'] = te['y_prob']
    out_pred['run_name'] = run_name
    out_pred['scale'] = scale
    out_pred['include_ma'] = include_ma
    out_pred['include_volume'] = include_volume

    ckpt_fp = MODEL_DIR / f'{run_name}.pt'
    torch.save(best_state, ckpt_fp)

    return {
        'run_name': run_name,
        'model_type': 'single_scale',
        'scale': scale,
        'include_ma': include_ma,
        'include_volume': include_volume,
        'best_epoch': best_epoch,
        'best_val_auc': best_auc,
        'threshold': best_thr,
        'val_metrics': val_metrics,
```

```
            'test_metrics': test_metrics,
            'history': pd.DataFrame(history),
            'test_pred': out_pred,
            'ckpt_fp': str(ckpt_fp),
            'model_obj': model,
        }

def train_multi_scale(
    manifest_df,
    scales,
    include_ma=True,
    include_volume=True,
    lr=1e-4,
    weight_decay=1e-4,
    max_epochs=20,
    patience=5,
    use_cache=False,
    run_name='multi',
):
    train_df = manifest_df[manifest_df['split'] == 'train'].copy()
    val_df = manifest_df[manifest_df['split'] == 'val'].copy()
    test_df = manifest_df[manifest_df['split'] == 'test'].copy()

    ds_train = MultiScaleChartDataset(train_df, cf, scales, include_ma, include
    ds_val = MultiScaleChartDataset(val_df, cf, scales, include_ma, include_vol
    ds_test = MultiScaleChartDataset(test_df, cf, scales, include_ma, include_v

    dl_train = DataLoader(ds_train, batch_size=BATCH_SIZE, shuffle=True, num_wo
    dl_val = DataLoader(ds_val, batch_size=BATCH_SIZE, shuffle=False, num_worke
    dl_test = DataLoader(ds_test, batch_size=BATCH_SIZE, shuffle=False, num_wor

    model = MultiScaleFusionCNN(scales).to(DEVICE)
    optimizer = torch.optim.AdamW(model.parameters(), lr=lr, weight_decay=weigh

    pos = train_df['label_up'].mean()
    w0 = 1.0 / max(1.0 - pos, 1e-6)
    w1 = 1.0 / max(pos, 1e-6)
    class_weights = torch.tensor([w0, w1], dtype=torch.float32, device=DEVICE)

    best_auc = -np.inf
    best_state = None
    best_epoch = -1
    bad = 0
    history = []

    for epoch in range(1, max_epochs + 1):
        tr = run_epoch_multi(model, dl_train, optimizer=optimizer, class_weight
        va = run_epoch_multi(model, dl_val, optimizer=None, class_weights=class

        tr_m = metric_bundle(tr['y_true'], tr['y_prob'], threshold=0.5)
        va_m = metric_bundle(va['y_true'], va['y_prob'], threshold=0.5)
```

```
        history.append({
            'epoch': epoch,
            'train_loss': tr['loss'],
            'val_loss': va['loss'],
            'train_auc': tr_m['roc_auc'],
            'val_auc': va_m['roc_auc'],
            'train_acc': tr_m['accuracy'],
            'val_acc': va_m['accuracy'],
        })

        if va_m['roc_auc'] > best_auc:
            best_auc = va_m['roc_auc']
            best_state = {k: v.detach().cpu() for k, v in model.state_dict().it
            best_epoch = epoch
            bad = 0
        else:
            bad += 1
            if bad >= patience:
                break

    model.load_state_dict(best_state)
    va = run_epoch_multi(model, dl_val, optimizer=None, class_weights=class_wei
    te = run_epoch_multi(model, dl_test, optimizer=None, class_weights=class_we

    best_thr, _ = choose_threshold(va['y_true'], va['y_prob'], metric=THRESHOLD
    val_metrics = metric_bundle(va['y_true'], va['y_prob'], threshold=best_thr)
    test_metrics = metric_bundle(te['y_true'], te['y_prob'], threshold=best_thr

    out_pred = test_df[['sample_id', 'anchor_ts', 'contract', 'forward_return',
    out_pred['prob_up'] = te['y_prob']
    out_pred['run_name'] = run_name
    out_pred['scales'] = str(list(scales))
    out_pred['include_ma'] = include_ma
    out_pred['include_volume'] = include_volume

    ckpt_fp = MODEL_DIR / f'{run_name}.pt'
    torch.save(best_state, ckpt_fp)

    return {
        'run_name': run_name,
        'model_type': 'multi_scale',
        'scales': list(scales),
        'include_ma': include_ma,
        'include_volume': include_volume,
        'best_epoch': best_epoch,
        'best_val_auc': best_auc,
        'threshold': best_thr,
        'val_metrics': val_metrics,
        'test_metrics': test_metrics,
        'history': pd.DataFrame(history),
        'test_pred': out_pred,
```

```
        'ckpt_fp': str(ckpt_fp),
        'model_obj': model,
    }
```

8) Simple benchmark comparisons

We train simple non-image baselines on numeric features from the same chronology-safe sample set.

```
def run_numeric_benchmarks(bench_df):
    feat_cols = ['mom_5h', 'mom_15h', 'range_20', 'close_vs_ma20', 'vol_ratio_2
    tr = bench_df[bench_df['split'] == 'train'].copy()
    va = bench_df[bench_df['split'] == 'val'].copy()
    te = bench_df[bench_df['split'] == 'test'].copy()

    # Logistic regression benchmark
    logit = LogisticRegression(max_iter=2000, class_weight='balanced')
    logit.fit(tr[feat_cols], tr['label_up'])
    va_prob = logit.predict_proba(va[feat_cols])[:, 1]
    te_prob = logit.predict_proba(te[feat_cols])[:, 1]
    thr, _ = choose_threshold(va['label_up'], va_prob, metric=THRESHOLD_METRIC)
    logit_test = metric_bundle(te['label_up'], te_prob, threshold=thr)

    # Momentum sign benchmark (very simple directional rule)
    te_simple_pred = (te['mom_5h'] > 0).astype(int).values
    simple = {
        'rows': len(te),
        'accuracy': accuracy_score(te['label_up'], te_simple_pred),
        'balanced_accuracy': balanced_accuracy_score(te['label_up'], te_simple_
        'precision': precision_score(te['label_up'], te_simple_pred, zero_divis
        'recall': recall_score(te['label_up'], te_simple_pred, zero_division=0)
        'f1': f1_score(te['label_up'], te_simple_pred, zero_division=0),
        'mcc': matthews_corrcoef(te['label_up'], te_simple_pred),
    }

    return {
        'logit_threshold': thr,
        'logit_test_metrics': logit_test,
        'simple_momentum_metrics': simple
    }

bench_results = run_numeric_benchmarks(bench_df)
bench_results
```

```
{'logit_threshold': 0.49499999999999994,
 'logit_test_metrics': {'rows': 2722,
  'threshold': 0.49499999999999994,
  'coverage': 1.0,
```

```
    'pred_positive_rate': 0.7538574577516532,
    'true_positive_rate': 0.4210139603232917,
    'accuracy': 0.45554739162380603,
    'balanced_accuracy': np.float64(0.4955384431392351),
    'precision': 0.41812865497076024,
    'recall': 0.7486910994764397,
    'f1': 0.5365853658536586,
    'mcc': np.float64(-0.01022729454476034),
    'roc_auc': np.float64(0.5228869340278701),
    'log_loss': 0.6927072561808082,
    'brier': np.float64(0.24978004166042933),
    'tn': 382,
    'fp': 1194,
    'fn': 288,
    'tp': 858},
   'simple_momentum_metrics': {'rows': 2722,
    'accuracy': 0.5216752387950037,
    'balanced_accuracy': np.float64(0.5183611502378611),
    'precision': 0.4398148148148148,
    'recall': 0.4973821989528796,
    'f1': 0.4668304668304668,
    'mcc': np.float64(0.036302625185435915)}}
```

```
# ============================================================
# HOTFIX: remove non-collatable Timestamp objects from batches
# ============================================================

class SingleScaleChartDataset(Dataset):
    def __init__(self, manifest_df, cf_df, scale, include_ma=True, include_volu
        self.df = manifest_df.reset_index(drop=True)
        self.cf = cf_df
        self.scale = scale
        self.include_ma = include_ma
        self.include_volume = include_volume
        self.use_cache = use_cache
        self.cache_dir = get_cache_subdir(scale, include_ma, include_volume)

    def __len__(self):
        return len(self.df)

    def __getitem__(self, idx):
        r = self.df.iloc[idx]
        sid = int(r['sample_id'])

        if self.use_cache and (self.cache_dir / f'{sid}.png').exists():
            img = Image.open(self.cache_dir / f'{sid}.png').convert('L')
        else:
            a = int(r['anchor_idx'])
            window = self.cf.iloc[a - self.scale + 1: a + 1].copy()
            img = render_refstyle_window(window, self.scale, self.include_ma, s

        x = to_tensor(img)
        y = torch.tensor(int(r['label_up']), dtype=torch.long)
```

```
        # Return only collatable items for DataLoader
        return {
            'x': x,
            'y': y,
            'sample_id': torch.tensor(sid, dtype=torch.long),
        }


class MultiScaleChartDataset(Dataset):
    def __init__(self, manifest_df, cf_df, scales, include_ma=True, include_vol
        self.df = manifest_df.reset_index(drop=True)
        self.cf = cf_df
        self.scales = list(scales)
        self.include_ma = include_ma
        self.include_volume = include_volume
        self.use_cache = use_cache
        self.cache_dirs = {s: get_cache_subdir(s, include_ma, include_volume) f

    def __len__(self):
        return len(self.df)

    def __getitem__(self, idx):
        r = self.df.iloc[idx]
        sid = int(r['sample_id'])
        out = {}

        for s in self.scales:
            fp = self.cache_dirs[s] / f'{sid}.png'
            if self.use_cache and fp.exists():
                img = Image.open(fp).convert('L')
            else:
                a = int(r['anchor_idx'])
                window = self.cf.iloc[a - s + 1: a + 1].copy()
                img = render_refstyle_window(window, s, self.include_ma, self.i
            out[f'x_{s}'] = to_tensor(img)

        out['y'] = torch.tensor(int(r['label_up']), dtype=torch.long)
        out['sample_id'] = torch.tensor(sid, dtype=torch.long)
        return out


# Safer for Colab debugging after class redefinition
NUM_WORKERS = 0

print("Dataset hotfix applied. NUM_WORKERS set to 0 for safe rerun.")
```

```
Dataset hotfix applied. NUM_WORKERS set to 0 for safe rerun.
```

9) Run the image-component ablations

This covers:

- OHLC only
- OHLC + MA
- OHLC + volume
- OHLC + MA + volume

First run the **single-scale I60** ablations.

```
RUN_ABLATIONS = True

ablation_runs = []
if RUN_ABLATIONS:
    for include_ma, include_volume in ABLATION_GRID:
        run_name = f"single_I60_ma{int(include_ma)}_vol{int(include_volume)}"
        print('\n====', run_name, '====')
        out = train_single_scale(
            manifest_df=master_q,
            scale=60,
            include_ma=include_ma,
            include_volume=include_volume,
            lr=LR_SINGLE,
            weight_decay=WEIGHT_DECAY,
            max_epochs=MAX_EPOCHS,
            patience=PATIENCE,
            use_cache=PRE_RENDER_IMAGES,
            run_name=run_name,
        )
        ablation_runs.append(out)

ablation_summary = []
for out in ablation_runs:
    row = {
        'run_name': out['run_name'],
        'scale': out['scale'],
        'include_ma': out['include_ma'],
        'include_volume': out['include_volume'],
        'best_epoch': out['best_epoch'],
        'best_val_auc': out['best_val_auc'],
        'threshold': out['threshold'],
    }
    row.update({f"test_{k}": v for k, v in out['test_metrics'].items() if k in
    ablation_summary.append(row)

ablation_summary_df = pd.DataFrame(ablation_summary).sort_values(['test_balance
```

```
ablation_summary_df.to_csv(RESULT_DIR / 'ablation_summary.csv', index=False)
ablation_summary_df
```

```
==== single_I60_ma0_vol0 ====
```

```
==== single_I60_ma1_vol0 ====
```

```
==== single_I60_ma0_vol1 ====
```

10) Train the multi-scale fusion model

This is the main model for the **multi-scale intraday input** point. Default scales are `20` and `60`.

```
RUN_MULTI_SCALE = True

multi_out = None
if RUN_MULTI_SCALE:
    run_name = f"multi_{'_'.join(map(str,SCALES))}_ma1_vol1"
    print('Running', run_name)
    multi_out = train_multi_scale(
        manifest_df=master_q,
        scales=SCALES,
        include_ma=True,
        include_volume=True,
        lr=LR_MULTI,
        weight_decay=WEIGHT_DECAY,
        max_epochs=MAX_EPOCHS,
        patience=PATIENCE,
        use_cache=PRE_RENDER_IMAGES,
        run_name=run_name,
    )

multi_summary_df = pd.DataFrame([{
    'run_name': multi_out['run_name'],
    'scales': str(multi_out['scales']),
    'best_epoch': multi_out['best_epoch'],
    'best_val_auc': multi_out['best_val_auc'],
    'threshold': multi_out['threshold'],
    **{f"test_{k}": v for k, v in multi_out['test_metrics'].items() if k in ['a
}]) if multi_out is not None else pd.DataFrame()

multi_summary_df.to_csv(RESULT_DIR / 'multi_scale_summary.csv', index=False)
multi_summary_df
```

	run_name	scale	include_ma	include_volume	best_epoch	best_val_
0	single_l60_ma0_vol0	60	False	False	5	0.569
2	single_l60_ma0_vol1	60	False	True	7	0.569
3	single_l60_ma1_vol1	60	True	True	3	0.554
1	single_l60_ma1_vol0	60	True	False	2	0.563

```
Running multi_20_60_ma1_vol1
```

	run_name	scales	best_epoch	best_val_auc	threshold	test_accura
0	multi_20_60_ma1_vol1	[20,	3	0.575133	0.46	0.4746

11) Tail-rule decision support

We evaluate **two-sided tail rules** on validation-calibrated probability rankings. This is the main operational decision-support layer.

```
def tail_grid_from_pred_df(pred_df, tail_grid=TAIL_GRID):
    y_true = pred_df['label_up'].values.astype(int)
    y_prob = pred_df['prob_up'].values.astype(float)
    rows = []
    for q in tail_grid:
        out = evaluate_tail_rule(y_true, y_prob, top_q=q, bottom_q=q)
```

```
            out['tail_q_each_side'] = q
            rows.append(out)
        return pd.DataFrame(rows)

tail_tables = []

# Best ablation single-scale
if len(ablation_summary_df):
    best_single_name = ablation_summary_df.iloc[0]['run_name']
    best_single = [x for x in ablation_runs if x['run_name'] == best_single_nam
    t1 = tail_grid_from_pred_df(best_single['test_pred'])
    t1['source'] = best_single_name
    tail_tables.append(t1)

# Multi-scale
if multi_out is not None:
    t2 = tail_grid_from_pred_df(multi_out['test_pred'])
    t2['source'] = multi_out['run_name']
    tail_tables.append(t2)

tail_grid_df = pd.concat(tail_tables, ignore_index=True) if tail_tables else pc
tail_grid_df.to_csv(RESULT_DIR / 'tail_grid_results.csv', index=False)
tail_grid_df.sort_values(['accuracy','balanced_accuracy','mcc'], ascending=Fals
```

	coverage	rows_kept	tail_low_thr	tail_high_thr	accuracy	balanced_accuracy
3	0.100661	274	0.383635	0.728554	0.693431	0.693596
2	0.050698	138	0.356887	0.760221	0.688406	0.695340
7	0.050698	138	0.404683	0.589725	0.688406	0.690365
6	0.030125	82	0.393569	0.596356	0.670732	0.670833
4	0.200588	546	0.414792	0.682294	0.666667	0.667965
5	0.020573	56	0.384274	0.599118	0.660714	0.660920
8	0.100661	274	0.420745	0.578357	0.649635	0.652263
1	0.030125	82	0.337683	0.772648	0.634146	0.637082
0	0.020573	56	0.325335	0.781543	0.607143	0.607692
9	0.200588	546	0.438455	0.562421	0.573260	0.573308

12) Pick the current best model for explainability analysis

We choose the strongest candidate by **tail-rule balanced accuracy**, then run Grad-CAM on the held-out test set.

```
# Pick best model source by tail balanced accuracy, then mcc, then coverage
assert len(tail_grid_df) > 0, 'Run model training first.'
best_tail_row = (
    tail_grid_df.sort_values(['balanced_accuracy','mcc','coverage'], ascending=
                .iloc[0]
)
best_source = best_tail_row['source']
best_tail_q = float(best_tail_row['tail_q_each_side'])
print('Best source:', best_source)
print('Best two-sided tail q:', best_tail_q)
print(best_tail_row)

if multi_out is not None and best_source == multi_out['run_name']:
    final_model = multi_out['model_obj']
    final_pred_df = multi_out['test_pred'].copy()
    final_model_type = 'multi'
else:
    final_model = [x for x in ablation_runs if x['run_name'] == best_source][0]
    final_pred_df = [x for x in ablation_runs if x['run_name'] == best_source][
    final_model_type = 'single'

final_model = final_model.to(DEVICE)
final_model.eval()
```

```
Best source: single_I60_ma0_vol0
Best two-sided tail q: 0.025
coverage                       0.050698
rows_kept                           138
tail_low_thr                   0.356887
tail_high_thr                  0.760221
accuracy                       0.688406
balanced_accuracy               0.69534
precision                      0.782609
recall                         0.658537
f1                             0.715232
mcc                            0.383683
tail_q_each_side                  0.025
source               single_I60_ma0_vol0
Name: 2, dtype: object
RefStyleCNN(
  (fe): CNNFeatureExtractor(
    (backbone): Sequential(
      (0): Conv2d(1, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      (1): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
      (2): LeakyReLU(negative_slope=0.01, inplace=True)
      (3): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
      (4): Conv2d(32, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      (5): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
```

```
      (6): LeakyReLU(negative_slope=0.01, inplace=True)
      (7): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
      (8): Conv2d(64, 128, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      (9): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
      (10): LeakyReLU(negative_slope=0.01, inplace=True)
      (11): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
      (12): Conv2d(128, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      (13): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
      (14): LeakyReLU(negative_slope=0.01, inplace=True)
      (15): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
    )
    (pool): AdaptiveAvgPool2d(output_size=(4, 4))
  )
  (head): Sequential(
    (0): Dropout(p=0.5, inplace=False)
    (1): Linear(in_features=4096, out_features=2, bias=True)
  )
)
```

13) Grad-CAM and chart-vs-background analysis

This explicitly addresses the supervisor request:

- quantify attention on **chart / foreground**
- quantify attention on **background**
- compare performance between those groups

```
def build_foreground_mask_from_image(img_pil: Image.Image):
    arr = np.array(img_pil)
    # any non-black pixel counts as foreground (OHLC + optional MA + optional v
    fg = (arr > 0).astype(np.uint8)
    return fg

def gradcam_single(model, x_tensor):
    # x_tensor shape [1,1,H,W]
    target_layer = None
    for m in model.modules():
        if isinstance(m, nn.Conv2d):
            target_layer = m
    activations = []
    gradients = []
```

```
        def fwd_hook(module, inp, out):
            activations.append(out)

        def bwd_hook(module, grad_in, grad_out):
            gradients.append(grad_out[0])

        h1 = target_layer.register_forward_hook(fwd_hook)
        h2 = target_layer.register_full_backward_hook(bwd_hook)

        logits = model(x_tensor)
        cls = logits[:, 1].sum()
        model.zero_grad(set_to_none=True)
        cls.backward()

        acts = activations[0].detach()
        grads_ = gradients[0].detach()

        weights = grads_.mean(dim=(2,3), keepdim=True)
        cam = (weights * acts).sum(dim=1, keepdim=True)
        cam = torch.relu(cam)
        cam = nn.functional.interpolate(cam, size=x_tensor.shape[-2:], mode='biline
        cam = cam[0,0].cpu().numpy()

        h1.remove(); h2.remove()

        cam = cam - cam.min()
        cam = cam / (cam.max() + 1e-9)
        return cam

    def gradcam_multi(model, xdict):
        # use the largest scale branch for localization
        largest = max(model.scales)
        target_branch = model.branches[str(largest)]
        target_layer = None
        for m in target_branch.modules():
            if isinstance(m, nn.Conv2d):
                target_layer = m
        activations = []
        gradients = []

        def fwd_hook(module, inp, out):
            activations.append(out)

        def bwd_hook(module, grad_in, grad_out):
            gradients.append(grad_out[0])

        h1 = target_layer.register_forward_hook(fwd_hook)
        h2 = target_layer.register_full_backward_hook(bwd_hook)

        logits = model(xdict)
        cls = logits[:, 1].sum()
```

```
        model.zero_grad(set_to_none=True)
        cls.backward()

        acts = activations[0].detach()
        grads_ = gradients[0].detach()
        weights = grads_.mean(dim=(2,3), keepdim=True)
        cam = (weights * acts).sum(dim=1, keepdim=True)
        cam = torch.relu(cam)

        H = xdict[f'x_{largest}'].shape[-2]
        W = xdict[f'x_{largest}'].shape[-1]
        cam = nn.functional.interpolate(cam, size=(H,W), mode='bilinear', align_cor
        cam = cam[0,0].cpu().numpy()

        h1.remove(); h2.remove()

        cam = cam - cam.min()
        cam = cam / (cam.max() + 1e-9)
        return cam, largest

    def evaluate_gradcam_groups(final_model, final_model_type, final_pred_df, tail_
        # choose tail-kept rows only for decision-support explainability
        y_prob = final_pred_df['prob_up'].values
        low_thr = np.quantile(y_prob, tail_q)
        high_thr = np.quantile(y_prob, 1 - tail_q)
        keep = (y_prob <= low_thr) | (y_prob >= high_thr)

        df = final_pred_df.loc[keep].copy().reset_index(drop=True)
        if max_eval is not None:
            df = df.head(max_eval).copy()

        rows = []
        for _, r in tqdm(df.iterrows(), total=len(df), desc='Grad-CAM evaluation'):
            sid = int(r['sample_id'])

            if final_model_type == 'single':
                scale = 60 if 'scale' not in r.index else int(r['scale'])
                # final source may be I60 ablation
                img = render_refstyle_window(
                    cf.iloc[int(master_q.loc[master_q['sample_id'] == sid, 'anchor_
                            int(master_q.loc[master_q['sample_id'] == sid, 'anchor_
                    scale=scale,
                    include_ma=bool(r.get('include_ma', True)),
                    include_volume=bool(r.get('include_volume', True)),
                )
                x = to_tensor(img).unsqueeze(0).to(DEVICE)
                cam = gradcam_single(final_model, x)
                img_arr = np.array(img)
            else:
                # multi-scale: localize on largest scale
                sample_row = master_q.loc[master_q['sample_id'] == sid].iloc[0]
```

```
                xdict = {}
                imgs = {}
                for s in SCALES:
                    img = render_refstyle_window(
                        cf.iloc[int(sample_row['anchor_idx']) - s + 1: int(sample_r
                        scale=s, include_ma=True, include_volume=True
                    )
                    imgs[s] = img
                    xdict[f'x_{s}'] = to_tensor(img).unsqueeze(0).to(DEVICE)
                cam, largest = gradcam_multi(final_model, xdict)
                img_arr = np.array(imgs[largest])

            fg_mask = build_foreground_mask_from_image(Image.fromarray(img_arr))
            bg_mask = 1 - fg_mask

            fg_heat = float((cam * fg_mask).sum() / (cam.sum() + 1e-9))
            bg_heat = float((cam * bg_mask).sum() / (cam.sum() + 1e-9))
            focus_group = 'chart_focused' if fg_heat >= bg_heat else 'background_fc

            pred = 1 if r['prob_up'] >= high_thr else 0
            correct = int(pred == int(r['label_up']))

            rows.append({
                'sample_id': sid,
                'prob_up': float(r['prob_up']),
                'label_up': int(r['label_up']),
                'pred_tail': int(pred),
                'correct_tail': correct,
                'forward_return': float(r['forward_return']),
                'foreground_heat': fg_heat,
                'background_heat': bg_heat,
                'focus_group': focus_group,
                'contract': r['contract'],
                'anchor_ts': r['anchor_ts'],
            })

        detail = pd.DataFrame(rows)
        summary = detail.groupby('focus_group').agg(
            rows=('sample_id','count'),
            accuracy=('correct_tail','mean'),
            avg_prob=('prob_up','mean'),
            avg_fg_heat=('foreground_heat','mean'),
            avg_bg_heat=('background_heat','mean'),
            avg_forward_return=('forward_return','mean'),
        ).reset_index()
        return detail, summary

    RUN_GRADCAM_ANALYSIS = True
    gradcam_detail_df = pd.DataFrame()
    gradcam_summary_df = pd.DataFrame()
```

```
if RUN_GRADCAM_ANALYSIS:
    gradcam_detail_df, gradcam_summary_df = evaluate_gradcam_groups(
        final_model=final_model,
        final_model_type=final_model_type,
        final_pred_df=final_pred_df,
        tail_q=best_tail_q,
        max_eval=None,     # set a cap if you want a faster first run
    )
    gradcam_detail_df.to_csv(RESULT_DIR / 'gradcam_chart_vs_background_detail.c
    gradcam_summary_df.to_csv(RESULT_DIR / 'gradcam_chart_vs_background_summary

gradcam_summary_df
```

```
Grad-                                                    138/138 [00:01<00:00, 116.30it/s]
CAM evaluation: 100%
```

focus_group	rows	accuracy	avg_prob	avg_fg_heat	avg_bg_heat	avg_fo

14) Final comparison tables and exports

1. List item
- List item
- List item
2. List item

This section writes clean CSV outputs for the thesis and supervisor update.

```
# Collect benchmark rows
final_rows = []

# Simple benchmarks
final_rows.append({
    'system': 'numeric_logit_benchmark',
    **{k: v for k, v in bench_results['logit_test_metrics'].items() if k in ['a
    'coverage': 1.0
})
final_rows.append({
    'system': 'simple_momentum_sign',
    **{k: v for k, v in bench_results['simple_momentum_metrics'].items() if k i
    'roc_auc': np.nan,
    'coverage': 1.0
})

# Best single-scale full-sample
if len(ablation_summary_df):
```

```
    best_single_name = ablation_summary_df.iloc[0]['run_name']
    best_single = [x for x in ablation_runs if x['run_name'] == best_single_nam
    final_rows.append({
        'system': f'best_single_fullsample::{best_single_name}',
        **{k: v for k, v in best_single['test_metrics'].items() if k in ['accur
        'coverage': 1.0
    })

# Best multi-scale full-sample
if multi_out is not None:
    final_rows.append({
        'system': f'multiscale_fullsample::{multi_out["run_name"]}',
        **{k: v for k, v in multi_out['test_metrics'].items() if k in ['accurac
        'coverage': 1.0
    })

# Tail-rule best
final_rows.append({
    'system': f'best_tail_rule::{best_source}::q={best_tail_q}',
    'accuracy': float(best_tail_row['accuracy']),
    'balanced_accuracy': float(best_tail_row['balanced_accuracy']),
    'precision': float(best_tail_row['precision']) if 'precision' in best_tail_
    'recall': float(best_tail_row['recall']) if 'recall' in best_tail_row else
    'f1': float(best_tail_row['f1']),
    'mcc': float(best_tail_row['mcc']),
    'roc_auc': np.nan,
    'coverage': float(best_tail_row['coverage']),
})

final_compare_df = pd.DataFrame(final_rows)
final_compare_df.to_csv(RESULT_DIR / 'final_compare_table.csv', index=False)
final_compare_df
```

	system	accuracy	balanced_accuracy	precisio
0	numeric_logit_benchmark	0.455547	0.495538	0.41812
1	simple_momentum_sign	0.521675	0.518361	0.43981
2	best_single_fullsample::single_l60_ma0_vol0	0.587068	0.559001	0.51291
3	multiscale_fullsample::multi_20_60_ma1_vol1	0.474651	0.520250	0.43358
4	best_tail_rule::single_l60_ma0_vol0::q=0.025	0.688406	0.695340	0.78260

```
# Save a compact "points covered" checklist
points_checklist = pd.DataFrame([
    {'point': 'Multi-scale intraday input', 'status': 'DONE' if multi_out is no
    {'point': 'Image-component ablations', 'status': 'DONE' if len(ablation_sum
    {'point': 'Selective tail-rule decision support', 'status': 'DONE' if len(t
    {'point': 'Grad-CAM chart-vs-background analysis', 'status': 'DONE' if len(
```

```
    {'point': 'Simple benchmark comparisons', 'status': 'DONE', 'evidence_file'
])
points_checklist.to_csv(RESULT_DIR / 'five_points_checklist.csv', index=False)
points_checklist
```

	point	status	evidence_file
0	Multi-scale intraday input	DONE	multi_scale_summary.csv
1	Image-component ablations	DONE	ablation_summary.csv
2	Selective tail-rule decision support	DONE	tail_grid_results.csv
3	Grad-CAM chart-vs-background analysis	DONE	gradcam_chart_vs_background_summary.csv

```
# ============================================================
# OPTION B — CELL 1
# Safety checks + upgraded config
# Run this first, then paste the output here.
# ============================================================

required_names = [
    'cf', 'build_master_manifest', 'render_refstyle_window',
    'metric_bundle', 'choose_threshold', 'evaluate_tail_rule',
    'RESULT_DIR', 'MODEL_DIR', 'CACHE_DIR', 'DERIVED_DIR',
    'BATCH_SIZE', 'MAX_EPOCHS', 'PATIENCE', 'DEVICE',
    'BAR_PIXELS', 'HEIGHT_MAP', 'PRICE_PANEL_RATIO',
    'VERTICAL_PADDING_FRAC', 'WEIGHT_DECAY', 'THRESHOLD_METRIC',
    'HORIZON', 'EXCLUDE_ROLL_DAY'
]
missing_required = [x for x in required_names if x not in globals()]
if missing_required:
    raise RuntimeError(f"Run the main workflow notebook first. Missing: {missir

# ----- Option B config -----
SCALES_B = [20, 60, 120]
MAX_LOOKBACK_B = max(SCALES_B)

REQUIRE_FUTURE_SAME_DAY_B = True   # keep output intraday
ALLOW_INPUT_CROSS_DAY_B = True     # needed for I120
USE_Q_FILTER_B = True
Q_FILTER_B = 0.10

# image height for I120
HEIGHT_MAP[120] = HEIGHT_MAP.get(120, 128)

# safer for Colab debugging
NUM_WORKERS_B = 0

print("Base workflow objects found.")
```

```
print("DEVICE:", DEVICE)
print("Option B scales:", SCALES_B)
print("MAX_LOOKBACK_B:", MAX_LOOKBACK_B)
print("REQUIRE_FUTURE_SAME_DAY_B:", REQUIRE_FUTURE_SAME_DAY_B)
print("ALLOW_INPUT_CROSS_DAY_B:", ALLOW_INPUT_CROSS_DAY_B)
print("Q_FILTER_B:", Q_FILTER_B)
print("HEIGHT_MAP[120]:", HEIGHT_MAP[120])
print("NUM_WORKERS_B:", NUM_WORKERS_B)
```

```
Base workflow objects found.
DEVICE: cuda
Option B scales: [20, 60, 120]
MAX_LOOKBACK_B: 120
REQUIRE_FUTURE_SAME_DAY_B: True
ALLOW_INPUT_CROSS_DAY_B: True
Q_FILTER_B: 0.1
HEIGHT_MAP[120]: 128
NUM_WORKERS_B: 0
```

```
# ============================================================
# OPTION B — CELL 2
# Rebuild master manifest for I120-capable experiment set
# Run this next, then paste the output here.
# ============================================================

master_b_fp = DERIVED_DIR / f'master_manifest_I{MAX_LOOKBACK_B}_R{HORIZON}_opti

if master_b_fp.exists():
    master_b = pd.read_parquet(master_b_fp)
else:
    master_b = build_master_manifest(
        cf=cf,
        max_lookback=MAX_LOOKBACK_B,
        horizon=HORIZON,
        require_future_same_day=REQUIRE_FUTURE_SAME_DAY_B,
        allow_input_cross_day=ALLOW_INPUT_CROSS_DAY_B,
        exclude_roll_day=EXCLUDE_ROLL_DAY,
    )
    master_b.to_parquet(master_b_fp, index=False)

# chronological split
dates_b = sorted(pd.Series(master_b['trade_date']).unique())
n_dates_b = len(dates_b)
train_end_b = int(0.70 * n_dates_b)
val_end_b = int(0.85 * n_dates_b)

train_dates_b = set(dates_b[:train_end_b])
val_dates_b = set(dates_b[train_end_b:val_end_b])
test_dates_b = set(dates_b[val_end_b:])

def assign_split_b(d):
```

```
        if d in train_dates_b:
            return 'train'
        if d in val_dates_b:
            return 'val'
        return 'test'

    master_b['split'] = master_b['trade_date'].map(assign_split_b)

    # q-filter from training only
    if USE_Q_FILTER_B:
        q_thr_b = master_b.loc[master_b['split'] == 'train', 'forward_return'].abs(
        master_b['keep_q'] = master_b['forward_return'].abs() > q_thr_b
    else:
        q_thr_b = 0.0
        master_b['keep_q'] = True

    master_b_q = master_b[master_b['keep_q']].copy().reset_index(drop=True)

    print("master_b shape:", master_b.shape)
    print("master_b_q shape:", master_b_q.shape)
    print("q-filter threshold:", q_thr_b)
    print("\nSplit counts:")
    print(master_b_q['split'].value_counts())

    print("\nPositive label rate by split:")
    display(master_b_q.groupby('split')['label_up'].mean().to_frame('positive_rate'

    print("\nContracts in master_b_q test split:")
    display(master_b_q.loc[master_b_q['split']=='test', 'contract'].value_counts().
```

```
Building master manifest: 100%                          29787/29787 [00:10<00:00, 3017.17it/s]
master_b shape: (18924, 12)
master_b_q shape: (17013, 12)
q-filter threshold: 0.0003875635173542502

Split counts:
split
train    11745
test      2687
val       2581
Name: count, dtype: int64

Positive label rate by split:
```

split	positive_rate
test	0.417566
train	0.480800
val	0.436652

```
Contracts in master_b_q test split:
```

contract	rows
SiU5	295
SiZ5	2392

```
# ============================================================
# OPTION B — CELL 3
# Collate-safe datasets for I120 and multi-scale runs
# Run this next, then paste the output here.
# ============================================================

from torchvision import transforms
to_tensor_b = transforms.Compose([transforms.ToTensor()])

class SingleScaleChartDatasetB(Dataset):
    def __init__(self, manifest_df, cf_df, scale, include_ma=True, include_volu
        self.df = manifest_df.reset_index(drop=True)
        self.cf = cf_df
        self.scale = scale
        self.include_ma = include_ma
        self.include_volume = include_volume

    def __len__(self):
        return len(self.df)
```

```
    def __getitem__(self, idx):
        r = self.df.iloc[idx]
        sid = int(r['sample_id'])
        a = int(r['anchor_idx'])
        window = self.cf.iloc[a - self.scale + 1: a + 1].copy()
        img = render_refstyle_window(window, self.scale, self.include_ma, self.
        x = to_tensor_b(img)
        y = torch.tensor(int(r['label_up']), dtype=torch.long)
        return {
            'x': x,
            'y': y,
            'sample_id': torch.tensor(sid, dtype=torch.long),
        }

class MultiScaleChartDatasetB(Dataset):
    def __init__(self, manifest_df, cf_df, scales, include_ma=True, include_vol
        self.df = manifest_df.reset_index(drop=True)
        self.cf = cf_df
        self.scales = list(scales)
        self.include_ma = include_ma
        self.include_volume = include_volume

    def __len__(self):
        return len(self.df)

    def __getitem__(self, idx):
        r = self.df.iloc[idx]
        sid = int(r['sample_id'])
        a = int(r['anchor_idx'])

        out = {}
        for s in self.scales:
            window = self.cf.iloc[a - s + 1: a + 1].copy()
            img = render_refstyle_window(window, s, self.include_ma, self.inclu
            out[f'x_{s}'] = to_tensor_b(img)

        out['y'] = torch.tensor(int(r['label_up']), dtype=torch.long)
        out['sample_id'] = torch.tensor(sid, dtype=torch.long)
        return out

# quick smoke test
tmp_single = SingleScaleChartDatasetB(master_b_q.head(2), cf, scale=120, incluc
tmp_multi = MultiScaleChartDatasetB(master_b_q.head(2), cf, scales=SCALES_B, ir

sample_single = tmp_single[0]
sample_multi = tmp_multi[0]

print("Single-scale sample keys:", sample_single.keys())
print("Single x shape:", sample_single['x'].shape, "| y:", sample_single['y'].i

print("Multi-scale sample keys:", sample_multi.keys())
```

```
for s in SCALES_B:
    print(f"x_{s} shape:", sample_multi[f'x_{s}'].shape)
print("Multi y:", sample_multi['y'].item())
```

```
Single-scale sample keys: dict_keys(['x', 'y', 'sample_id'])
Single x shape: torch.Size([1, 128, 360]) | y: 0
Multi-scale sample keys: dict_keys(['x_20', 'x_60', 'x_120', 'y', 'sample_id'])
x_20 shape: torch.Size([1, 64, 60])
x_60 shape: torch.Size([1, 96, 180])
x_120 shape: torch.Size([1, 128, 360])
Multi y: 0
```

```
# ============================================================
# OPTION B — CELL 4
# Better fusion model (gated fusion, stronger than simple concat)
# Run this next, then paste the output here.
# ============================================================

class CNNFeatureExtractorB(nn.Module):
    def __init__(self, in_channels=1, base=32, depth=4):
        super().__init__()
        channels = []
        c = base
        for _ in range(depth):
            channels.append(c)
            c = min(c * 2, 256)

        blocks = []
        prev = in_channels
        for c in channels:
            blocks.extend([
                nn.Conv2d(prev, c, kernel_size=3, padding=1, bias=False),
                nn.BatchNorm2d(c),
                nn.LeakyReLU(0.01, inplace=True),
                nn.MaxPool2d(2, 2),
            ])
            prev = c

        self.backbone = nn.Sequential(*blocks)
        self.pool = nn.AdaptiveAvgPool2d((4, 4))
        self.out_dim = prev * 4 * 4

    def forward(self, x):
        z = self.backbone(x)
        z = self.pool(z)
        return torch.flatten(z, 1)

def depth_for_scale_b(scale):
    if scale <= 20:
        return 3
    elif scale <= 60:
```

```
            return 4
        else:
            return 5

class RefStyleCNNB(nn.Module):
    def __init__(self, scale, dropout=0.5):
        super().__init__()
        self.scale = scale
        self.fe = CNNFeatureExtractorB(depth=depth_for_scale_b(scale))
        self.head = nn.Sequential(
            nn.Dropout(dropout),
            nn.Linear(self.fe.out_dim, 2),
        )

    def forward(self, x):
        z = self.fe(x)
        return self.head(z)

class MultiScaleGatedFusionCNN(nn.Module):
    def __init__(self, scales=(20, 60, 120), proj_dim=128, dropout=0.5):
        super().__init__()
        self.scales = list(scales)

        self.branches = nn.ModuleDict({
            str(s): CNNFeatureExtractorB(depth=depth_for_scale_b(s)) for s in s
        })

        self.proj = nn.ModuleDict({
            str(s): nn.Sequential(
                nn.Linear(self.branches[str(s)].out_dim, proj_dim),
                nn.LayerNorm(proj_dim),
                nn.LeakyReLU(0.01, inplace=True),
            ) for s in self.scales
        })

        self.gate_net = nn.Sequential(
            nn.Linear(len(self.scales) * proj_dim, 128),
            nn.LeakyReLU(0.01, inplace=True),
            nn.Dropout(dropout),
            nn.Linear(128, len(self.scales))
        )

        self.head = nn.Sequential(
            nn.Dropout(dropout),
            nn.Linear(proj_dim, 64),
            nn.LeakyReLU(0.01, inplace=True),
            nn.Dropout(dropout),
            nn.Linear(64, 2)
        )

    def forward(self, batch_x_dict):
```

```
        proj_feats = []
        for s in self.scales:
            z = self.branches[str(s)](batch_x_dict[f'x_{s}'])
            z = self.proj[str(s)](z)
            proj_feats.append(z)

        cat = torch.cat(proj_feats, dim=1)
        gate_logits = self.gate_net(cat)
        gates = torch.softmax(gate_logits, dim=1)

        fused = 0
        for i, z in enumerate(proj_feats):
            fused = fused + gates[:, i:i+1] * z

        return self.head(fused)

# quick smoke test
single_model_b = RefStyleCNNB(scale=120).to(DEVICE)
multi_model_b = MultiScaleGatedFusionCNN(scales=SCALES_B).to(DEVICE)

x_single = sample_single['x'].unsqueeze(0).to(DEVICE)  # [1,1,H,W]
x_multi = {f'x_{s}': sample_multi[f'x_{s}'].unsqueeze(0).to(DEVICE) for s in SC

with torch.no_grad():
    logits_single = single_model_b(x_single)
    logits_multi = multi_model_b(x_multi)

print("Single model logits shape:", logits_single.shape)
print("Multi model logits shape:", logits_multi.shape)
print("Single feature dim:", single_model_b.fe.out_dim)
print("Multi branch dims:", {s: multi_model_b.branches[str(s)].out_dim for s in
print("Models ready.")
```

```
Single model logits shape: torch.Size([1, 2])
Multi model logits shape: torch.Size([1, 2])
Single feature dim: 4096
Multi branch dims: {20: 2048, 60: 4096, 120: 4096}
Models ready.
```

```
# ============================================================
# OPTION B — CELL 5
# Epoch runners for single-scale and gated multi-scale models
# Run this next, then paste the output here.
# ============================================================

def metric_bundle_b(y_true, y_prob, threshold=0.5):
    return metric_bundle(y_true, y_prob, threshold)

def choose_threshold_b(y_true, y_prob, metric='balanced_accuracy'):
    return choose_threshold(y_true, y_prob, metric)
```

```
def run_epoch_single_b(model, loader, optimizer=None, class_weights=None):
    train_mode = optimizer is not None
    model.train(train_mode)
    criterion = nn.CrossEntropyLoss(weight=class_weights)

    losses, probs, ys, ids = [], [], [], []

    for batch in tqdm(loader, leave=False):
        x = batch['x'].to(DEVICE, non_blocking=True)
        y = batch['y'].to(DEVICE, non_blocking=True)

        with torch.set_grad_enabled(train_mode):
            logits = model(x)
            loss = criterion(logits, y)
            if train_mode:
                optimizer.zero_grad(set_to_none=True)
                loss.backward()
                optimizer.step()

        p = torch.softmax(logits, dim=1)[:, 1].detach().cpu().numpy()
        losses.append(float(loss.item()))
        probs.extend(p.tolist())
        ys.extend(batch['y'].cpu().numpy().tolist())
        ids.extend(batch['sample_id'].cpu().numpy().tolist())

    return {
        'loss': float(np.mean(losses)),
        'y_true': np.array(ys),
        'y_prob': np.array(probs),
        'sample_id': np.array(ids)
    }

def run_epoch_multi_b(model, loader, optimizer=None, class_weights=None):
    train_mode = optimizer is not None
    model.train(train_mode)
    criterion = nn.CrossEntropyLoss(weight=class_weights)

    losses, probs, ys, ids = [], [], [], []

    for batch in tqdm(loader, leave=False):
        xdict = {k: v.to(DEVICE, non_blocking=True) for k, v in batch.items() i
        y = batch['y'].to(DEVICE, non_blocking=True)

        with torch.set_grad_enabled(train_mode):
            logits = model(xdict)
            loss = criterion(logits, y)
            if train_mode:
                optimizer.zero_grad(set_to_none=True)
                loss.backward()
                optimizer.step()
```

```
            p = torch.softmax(logits, dim=1)[:, 1].detach().cpu().numpy()
            losses.append(float(loss.item()))
            probs.extend(p.tolist())
            ys.extend(batch['y'].cpu().numpy().tolist())
            ids.extend(batch['sample_id'].cpu().numpy().tolist())

        return {
            'loss': float(np.mean(losses)),
            'y_true': np.array(ys),
            'y_prob': np.array(probs),
            'sample_id': np.array(ids)
        }

# quick smoke test with tiny loaders
tiny_train_single = DataLoader(
    SingleScaleChartDatasetB(master_b_q.head(8), cf, scale=120, include_ma=Fals
    batch_size=4, shuffle=False, num_workers=0
)
tiny_train_multi = DataLoader(
    MultiScaleChartDatasetB(master_b_q.head(8), cf, scales=SCALES_B, include_ma
    batch_size=4, shuffle=False, num_workers=0
)

cw = torch.tensor([1.0, 1.0], dtype=torch.float32, device=DEVICE)

single_test_out = run_epoch_single_b(single_model_b, tiny_train_single, optimiz
multi_test_out = run_epoch_multi_b(multi_model_b, tiny_train_multi, optimizer=N

print("Single runner OK:", single_test_out['y_prob'][:3], "| rows:", len(single
print("Multi runner OK:", multi_test_out['y_prob'][:3], "| rows:", len(multi_te
print("Epoch runners ready.")
```

```
Single runner OK: [0.49359488 0.49350202 0.4941453 ] | rows: 8
Multi runner OK: [0.50518078 0.5071736  0.51039845] | rows: 8
Epoch runners ready.
```

```
# ============================================================
# OPTION B — CELL 6
# Training helpers:
#   - single-scale I120
#   - gated multi-scale 20+60+120
# Run this next, then paste the output here.
# ============================================================

def train_single_scale_b(
    manifest_df,
    scale,
```

```
        include_ma=True,
        include_volume=True,
        lr=1e-4,
        weight_decay=1e-4,
        max_epochs=20,
        patience=5,
        run_name='single_b'
    ):
        train_df = manifest_df[manifest_df['split'] == 'train'].copy()
        val_df   = manifest_df[manifest_df['split'] == 'val'].copy()
        test_df  = manifest_df[manifest_df['split'] == 'test'].copy()

        ds_train = SingleScaleChartDatasetB(train_df, cf, scale, include_ma, incluc
        ds_val   = SingleScaleChartDatasetB(val_df, cf, scale, include_ma, include_
        ds_test  = SingleScaleChartDatasetB(test_df, cf, scale, include_ma, include

        dl_train = DataLoader(ds_train, batch_size=BATCH_SIZE, shuffle=True,  num_w
        dl_val   = DataLoader(ds_val,   batch_size=BATCH_SIZE, shuffle=False, num_w
        dl_test  = DataLoader(ds_test,  batch_size=BATCH_SIZE, shuffle=False, num_w

        model = RefStyleCNNB(scale=scale).to(DEVICE)
        optimizer = torch.optim.AdamW(model.parameters(), lr=lr, weight_decay=weigh

        pos = train_df['label_up'].mean()
        w0 = 1.0 / max(1.0 - pos, 1e-6)
        w1 = 1.0 / max(pos, 1e-6)
        class_weights = torch.tensor([w0, w1], dtype=torch.float32, device=DEVICE)

        best_auc = -np.inf
        best_state = None
        best_epoch = -1
        bad = 0
        history = []

        for epoch in range(1, max_epochs + 1):
            tr = run_epoch_single_b(model, dl_train, optimizer=optimizer, class_wei
            va = run_epoch_single_b(model, dl_val, optimizer=None, class_weights=cl

            tr_m = metric_bundle_b(tr['y_true'], tr['y_prob'], threshold=0.5)
            va_m = metric_bundle_b(va['y_true'], va['y_prob'], threshold=0.5)

            history.append({
                'epoch': epoch,
                'train_loss': tr['loss'],
                'val_loss': va['loss'],
                'train_auc': tr_m['roc_auc'],
                'val_auc': va_m['roc_auc'],
                'train_acc': tr_m['accuracy'],
                'val_acc': va_m['accuracy'],
            })
```

```
            print(f"[{run_name}] epoch {epoch:02d} | train_auc={tr_m['roc_auc']:.4f

            if va_m['roc_auc'] > best_auc:
                best_auc = va_m['roc_auc']
                best_state = {k: v.detach().cpu() for k, v in model.state_dict().it
                best_epoch = epoch
                bad = 0
            else:
                bad += 1
                if bad >= patience:
                    print(f"[{run_name}] early stopping at epoch {epoch}")
                    break

        model.load_state_dict(best_state)

        va = run_epoch_single_b(model, dl_val, optimizer=None, class_weights=class_
        te = run_epoch_single_b(model, dl_test, optimizer=None, class_weights=class

        best_thr, _ = choose_threshold_b(va['y_true'], va['y_prob'], metric=THRESHO
        val_metrics  = metric_bundle_b(va['y_true'], va['y_prob'], threshold=best_t
        test_metrics = metric_bundle_b(te['y_true'], te['y_prob'], threshold=best_t

        out_pred = test_df[['sample_id', 'anchor_ts', 'contract', 'forward_return',
        out_pred['prob_up'] = te['y_prob']
        out_pred['run_name'] = run_name
        out_pred['scale'] = scale
        out_pred['include_ma'] = include_ma
        out_pred['include_volume'] = include_volume

        ckpt_fp = MODEL_DIR / f'{run_name}.pt'
        torch.save(best_state, ckpt_fp)

        return {
            'run_name': run_name,
            'model_type': 'single_scale',
            'scale': scale,
            'include_ma': include_ma,
            'include_volume': include_volume,
            'best_epoch': best_epoch,
            'best_val_auc': best_auc,
            'threshold': best_thr,
            'val_metrics': val_metrics,
            'test_metrics': test_metrics,
            'history': pd.DataFrame(history),
            'test_pred': out_pred,
            'ckpt_fp': str(ckpt_fp),
            'model_obj': model,
        }

    def train_multi_scale_gated_b(
        manifest_df,
```

```
        scales=(20, 60, 120),
        include_ma=True,
        include_volume=True,
        lr=1e-4,
        weight_decay=1e-4,
        max_epochs=20,
        patience=5,
        run_name='multi_gated_b'
    ):
        train_df = manifest_df[manifest_df['split'] == 'train'].copy()
        val_df   = manifest_df[manifest_df['split'] == 'val'].copy()
        test_df  = manifest_df[manifest_df['split'] == 'test'].copy()

        ds_train = MultiScaleChartDatasetB(train_df, cf, scales, include_ma, incluc
        ds_val   = MultiScaleChartDatasetB(val_df, cf, scales, include_ma, include_
        ds_test  = MultiScaleChartDatasetB(test_df, cf, scales, include_ma, include

        dl_train = DataLoader(ds_train, batch_size=BATCH_SIZE, shuffle=True,  num_w
        dl_val   = DataLoader(ds_val,   batch_size=BATCH_SIZE, shuffle=False, num_w
        dl_test  = DataLoader(ds_test,  batch_size=BATCH_SIZE, shuffle=False, num_w

        model = MultiScaleGatedFusionCNN(scales=scales).to(DEVICE)
        optimizer = torch.optim.AdamW(model.parameters(), lr=lr, weight_decay=weigh

        pos = train_df['label_up'].mean()
        w0 = 1.0 / max(1.0 - pos, 1e-6)
        w1 = 1.0 / max(pos, 1e-6)
        class_weights = torch.tensor([w0, w1], dtype=torch.float32, device=DEVICE)

        best_auc = -np.inf
        best_state = None
        best_epoch = -1
        bad = 0
        history = []

        for epoch in range(1, max_epochs + 1):
            tr = run_epoch_multi_b(model, dl_train, optimizer=optimizer, class_weig
            va = run_epoch_multi_b(model, dl_val, optimizer=None, class_weights=cla

            tr_m = metric_bundle_b(tr['y_true'], tr['y_prob'], threshold=0.5)
            va_m = metric_bundle_b(va['y_true'], va['y_prob'], threshold=0.5)

            history.append({
                'epoch': epoch,
                'train_loss': tr['loss'],
                'val_loss': va['loss'],
                'train_auc': tr_m['roc_auc'],
                'val_auc': va_m['roc_auc'],
                'train_acc': tr_m['accuracy'],
                'val_acc': va_m['accuracy'],
            })
```

```
            print(f"[{run_name}] epoch {epoch:02d} | train_auc={tr_m['roc_auc']:.4f

            if va_m['roc_auc'] > best_auc:
                best_auc = va_m['roc_auc']
                best_state = {k: v.detach().cpu() for k, v in model.state_dict().it
                best_epoch = epoch
                bad = 0
            else:
                bad += 1
                if bad >= patience:
                    print(f"[{run_name}] early stopping at epoch {epoch}")
                    break

        model.load_state_dict(best_state)

        va = run_epoch_multi_b(model, dl_val, optimizer=None, class_weights=class_w
        te = run_epoch_multi_b(model, dl_test, optimizer=None, class_weights=class_

        best_thr, _ = choose_threshold_b(va['y_true'], va['y_prob'], metric=THRESHO
        val_metrics  = metric_bundle_b(va['y_true'], va['y_prob'], threshold=best_t
        test_metrics = metric_bundle_b(te['y_true'], te['y_prob'], threshold=best_t

        out_pred = test_df[['sample_id', 'anchor_ts', 'contract', 'forward_return',
        out_pred['prob_up'] = te['y_prob']
        out_pred['run_name'] = run_name
        out_pred['scales'] = str(list(scales))
        out_pred['include_ma'] = include_ma
        out_pred['include_volume'] = include_volume

        ckpt_fp = MODEL_DIR / f'{run_name}.pt'
        torch.save(best_state, ckpt_fp)

        return {
            'run_name': run_name,
            'model_type': 'multi_scale_gated',
            'scales': list(scales),
            'include_ma': include_ma,
            'include_volume': include_volume,
            'best_epoch': best_epoch,
            'best_val_auc': best_auc,
            'threshold': best_thr,
            'val_metrics': val_metrics,
            'test_metrics': test_metrics,
            'history': pd.DataFrame(history),
            'test_pred': out_pred,
            'ckpt_fp': str(ckpt_fp),
            'model_obj': model,
        }

    print("Option B training helpers ready.")
```

```
Option B training helpers ready.
```

```
# ============================================================
# OPTION B — CELL 7A
# Run single-scale I120, OHLC-only
# Run this next, then paste the final printed metrics here.
# ============================================================

run_I120_ohcl = train_single_scale_b(
    manifest_df=master_b_q,
    scale=120,
    include_ma=False,
    include_volume=False,
    lr=1e-4,
    weight_decay=WEIGHT_DECAY,
    max_epochs=MAX_EPOCHS,
    patience=PATIENCE,
    run_name='single_I120_ma0_vol0'
)

print("\n=== FINAL TEST METRICS: single_I120_ma0_vol0 ===")
for k, v in run_I120_ohcl['test_metrics'].items():
    print(f"{k}: {v}")
print("best_epoch:", run_I120_ohcl['best_epoch'])
print("best_val_auc:", run_I120_ohcl['best_val_auc'])
print("threshold:", run_I120_ohcl['threshold'])
```

```
[single_I120_ma0_vol0] epoch 01 | train_auc=0.5704 | val_auc=0.5699 | val_acc=

[single_I120_ma0_vol0] epoch 02 | train_auc=0.6187 | val_auc=0.5185 | val_acc=

[single_I120_ma0_vol0] epoch 03 | train_auc=0.6743 | val_auc=0.5190 | val_acc=
```

```
# ============================================================
# OPTION B — CELL 7B
# Run better gated fusion: 20 + 60 + 120, OHLC-only
# Run this next, then paste the final printed metrics here.
# ============================================================

run_multi_gated_ohcl = train_multi_scale_gated_b(
    manifest_df=master_b_q,
    scales=SCALES_B,
    include_ma=False,
    include_volume=False,
    lr=1e-4,
    weight_decay=WEIGHT_DECAY,
    max_epochs=MAX_EPOCHS,
    patience=PATIENCE,
    run_name='multi_gated_20_60_120_ma0_vol0'
)

print("\n=== FINAL TEST METRICS: multi_gated_20_60_120_ma0_vol0 ===")
for k, v in run_multi_gated_ohcl['test_metrics'].items():
    print(f"{k}: {v}")
print("best_epoch:", run_multi_gated_ohcl['best_epoch'])
print("best_val_auc:", run_multi_gated_ohcl['best_val_auc'])
print("threshold:", run_multi_gated_ohcl['threshold'])
```

```
[single_I120_ma0_vol0] epoch 09 | train_auc=0.9255 | val_auc=0.4786 | val_acc=
[single_I120_ma0_vol0] early stopping at epoch 9

=== FINAL TEST METRICS: single_I120_ma0_vol0 ===
rows: 2687
threshold: 0.5249999999999999
coverage: 1.0
pred_positive_rate: 0.2214365463342017
true_positive_rate: 0.4175660588016375
accuracy: 0.5686639374767398
balanced_accuracy: 0.523373084348465
```

```
precision: 0.4689075630252101
recall: 0.24866310160427807
f1: 0.32498543972044264
mcc: 0.05552138592609231
roc_auc: 0.5136594283371205
log_loss: 0.7535422175432587
brier: 0.27091897533126563
tn: 1249
fp: 316
fn: 843
tp: 279
best_epoch: 4
best_val_auc: 0.5769437551947996
threshold: 0.5249999999999999
```

```
[multi_gated_20_60_120_ma0_vol0] epoch 01 | train_auc=0.5380 | val_auc=0.5533
```

```
# ============================================================
# OPTION B — CELL 7C
# Run better gated fusion: 20 + 60 + 120, full image (MA + volume)
# This is the closest Option B test to the richer reference-paper logic.
# Run this next, then paste the final printed metrics here.
# ============================================================

run_multi_gated_full = train_multi_scale_gated_b(
    manifest_df=master_b_q,
    scales=SCALES_B,
    include_ma=True,
    include_volume=True,
    lr=1e-4,
    weight_decay=WEIGHT_DECAY,
    max_epochs=MAX_EPOCHS,
    patience=PATIENCE,
    run_name='multi_gated_20_60_120_ma1_vol1'
)

print("\n=== FINAL TEST METRICS: multi_gated_20_60_120_ma1_vol1 ===")
for k, v in run_multi_gated_full['test_metrics'].items():
    print(f"{k}: {v}")
print("best_epoch:", run_multi_gated_full['best_epoch'])
print("best_val_auc:", run_multi_gated_full['best_val_auc'])
print("threshold:", run_multi_gated_full['threshold'])
```

```
[multi_gated_20_60_120_ma0_vol0] early stopping at epoch 7
```

```
=== FINAL TEST METRICS: multi_gated_20_60_120_ma0_vol0 ===
rows: 2687
threshold: 0.4649999999999999
coverage: 1.0
pred_positive_rate: 0.5906215109787868
true_positive_rate: 0.4175660588016375
accuracy: 0.5039077037588389
balanced_accuracy: 0.5193749181345497
precision: 0.4335223692501575
recall: 0.6131907308377896
f1: 0.5079365079365079
mcc: 0.03886320963433026
roc_auc: 0.5585803534309455
log_loss: 0.6797808094492669
brier: 0.24335308712542644
tn: 666
fp: 899
fn: 434
```

```
tp: 688
best_epoch: 2
best_val_auc: 0.5640176290598771
threshold: 0.4649999999999999
```

```
[multi_gated_20_60_120_ma1_vol1] epoch 01 | train_auc=0.5307 | val_auc=0.5501



[multi_gated_20_60_120_ma1_vol1] epoch 02 | train_auc=0.5570 | val_auc=0.5635
```

```
# Put the models you already trained into one list
option_b_runs = [
    run_I120_ohcl,
    run_multi_gated_ohcl,
    run_multi_gated_full,
]

option_b_tail_rows = []

for out in option_b_runs:
    pred_df = out['test_pred'].copy()
    y_true = pred_df['label_up'].values.astype(int)
    y_prob = pred_df['prob_up'].values.astype(float)

    for q in [0.01, 0.015, 0.025, 0.05, 0.10]:
        res = evaluate_tail_rule(y_true, y_prob, top_q=q, bottom_q=q)
        res['tail_q_each_side'] = q
        res['source'] = out['run_name']
        option_b_tail_rows.append(res)

option_b_tail_df = pd.DataFrame(option_b_tail_rows).sort_values(
    ['balanced_accuracy', 'mcc', 'coverage'],
    ascending=False
).reset_index(drop=True)

option_b_tail_df.to_csv(RESULT_DIR / 'optionB_tail_grid_results.csv', index=Fal

print("Top Option B tail-rule results:")
display(option_b_tail_df.head(20))
```

```
=== FINAL TEST METRICS: multi_gated_20_60_120_ma1_vol1 ===
rows: 2687
threshold: 0.5049999999999999
coverage: 1.0
pred_positive_rate: 0.5262374395236323
true_positive_rate: 0.4175660588016375
accuracy: 0.540007443245255
balanced_accuracy: 0.5455718622040742
precision: 0.4596888260254597
recall: 0.5793226381461676
f1: 0.5126182965299685
mcc: 0.09002050896108849
roc_auc: 0.5486135552100596
```

```
log_loss: 0.6912824060899858
brier: 0.2490874694062442
tn: 801

tp: 650
best_epoch: 3
```

Top-Option B tail-rule results:

	coverage	rows_kept	tail_low_thr	tail_high_thr	accuracy	balanced_accurac
0	0.020097	54	0.081354	0.866855	0.833333	0.83379
1	0.020097	54	0.383513	0.577567	0.796296	0.80681
2	0.020097	54	0.287342	0.601700	0.777778	0.79220
3	0.030517	82	0.289713	0.598831	0.780488	0.78662
4	0.030517	82	0.089618	0.847163	0.780488	0.78318
5	0.050614	136	0.099133	0.810215	0.742647	0.74396
6	0.050614	136	0.295721	0.595308	0.727941	0.72799
7	0.030517	82	0.385712	0.570816	0.719512	0.72282
8	0.100484	270	0.405580	0.548964	0.718519	0.71972
9	0.050614	136	0.387824	0.563067	0.713235	0.71439
10	0.200223	538	0.421746	0.534497	0.620818	0.62098
11	0.100484	270	0.122038	0.734934	0.618519	0.61904
12	0.200223	538	0.397508	0.573242	0.585502	0.58613
13	0.100484	270	0.342923	0.585622	0.581481	0.58224
14	0.200223	538	0.155919	0.661642	0.559480	0.55977

```
# ============================================================
# OPTION B — TAIL CELL 2
# Build shortlist of the key finalists
# Run this next, then paste the table here.
# ============================================================

strict_best_row = option_b_tail_df.iloc[0].copy()

best_multiscale_row = (
    option_b_tail_df[option_b_tail_df['source'].str.contains('multi_gated')]
    .sort_values(['balanced_accuracy', 'mcc', 'coverage'], ascending=False)
    .iloc[0]
    .copy()
)

best_rich_row = (
    option_b_tail_df[option_b_tail_df['source'] == 'multi_gated_20_60_120_ma1_v
    .sort_values(['balanced_accuracy', 'mcc', 'coverage'], ascending=False)
    .iloc[0]
    .copy()
)
```

```
finalists_df = pd.DataFrame([
    {
        'role': 'strict_best_score',
        **strict_best_row.to_dict()
    },
    {
        'role': 'best_multiscale',
        **best_multiscale_row.to_dict()
    },
    {
        'role': 'best_rich_fullimage',
        **best_rich_row.to_dict()
    },
])

finalists_df.to_csv(RESULT_DIR / 'optionB_finalists.csv', index=False)
display(finalists_df)
```

	role	coverage	rows_kept	tail_low_thr	tail_high_thr	accuracy
0	strict_best_score	0.020097	54	0.081354	0.866855	0.833333
1	best_multiscale	0.020097	54	0.383513	0.577567	0.796296
2	best_rich_fullimage	0.020097	54	0.287342	0.601700	0.777778

```
# ============================================================
# OPTION B — TAIL CELL 3
# Define improved Grad-CAM + structured foreground masks
# Run this first, then paste the output here.
# ============================================================

from PIL import ImageFilter

def build_structured_masks_from_image(img_pil, price_panel_ratio=PRICE_PANEL_RA
    arr = np.array(img_pil)
    H, W = arr.shape
    price_h = int(round(H * price_panel_ratio))

    # exact visible foreground
    fg_exact = (arr > 0).astype(np.uint8)

    # dilated foreground so blurry CAM heat near candles is not counted as back
    fg_img = Image.fromarray((fg_exact * 255).astype(np.uint8))
    fg_dil = np.array(fg_img.filter(ImageFilter.MaxFilter(size=7)))
    fg_dil = (fg_dil > 0).astype(np.uint8)

    price_mask = np.zeros_like(arr, dtype=np.uint8)
    price_mask[:price_h, :] = 1

    volume_mask = np.zeros_like(arr, dtype=np.uint8)
```

```
        volume_mask[price_h:, :] = 1

        price_fg = fg_dil * price_mask
        volume_fg = fg_dil * volume_mask
        bg_mask = 1 - fg_dil

        # recent right-edge zone
        recent_mask = np.zeros_like(arr, dtype=np.uint8)
        recent_mask[:, max(0, W - 15):] = 1

        return {
            'fg_exact': fg_exact,
            'fg_dil': fg_dil,
            'price_mask': price_mask,
            'volume_mask': volume_mask,
            'price_fg': price_fg,
            'volume_fg': volume_fg,
            'bg_mask': bg_mask,
            'recent_mask': recent_mask,
        }

    def gradcam_single_b(model, x_tensor):
        target_layer = None
        for m in model.modules():
            if isinstance(m, nn.Conv2d):
                target_layer = m

        activations, gradients = [], []

        def fwd_hook(module, inp, out):
            activations.append(out)

        def bwd_hook(module, grad_in, grad_out):
            gradients.append(grad_out[0])

        h1 = target_layer.register_forward_hook(fwd_hook)
        h2 = target_layer.register_full_backward_hook(bwd_hook)

        logits = model(x_tensor)
        cls = logits[:, 1].sum()
        model.zero_grad(set_to_none=True)
        cls.backward()

        acts = activations[0].detach()
        grads_ = gradients[0].detach()
        weights = grads_.mean(dim=(2, 3), keepdim=True)
        cam = (weights * acts).sum(dim=1, keepdim=True)
        cam = torch.relu(cam)
        cam = nn.functional.interpolate(cam, size=x_tensor.shape[-2:], mode='biline
        cam = cam[0, 0].detach().cpu().numpy()
```

```
        h1.remove()
        h2.remove()

        cam = cam - cam.min()
        cam = cam / (cam.max() + 1e-9)
        return cam

    def gradcam_multi_b(model, xdict):
        largest = max(model.scales)
        target_branch = model.branches[str(largest)]

        target_layer = None
        for m in target_branch.modules():
            if isinstance(m, nn.Conv2d):
                target_layer = m

        activations, gradients = [], []

        def fwd_hook(module, inp, out):
            activations.append(out)

        def bwd_hook(module, grad_in, grad_out):
            gradients.append(grad_out[0])

        h1 = target_layer.register_forward_hook(fwd_hook)
        h2 = target_layer.register_full_backward_hook(bwd_hook)

        logits = model(xdict)
        cls = logits[:, 1].sum()
        model.zero_grad(set_to_none=True)
        cls.backward()

        acts = activations[0].detach()
        grads_ = gradients[0].detach()
        weights = grads_.mean(dim=(2, 3), keepdim=True)
        cam = (weights * acts).sum(dim=1, keepdim=True)
        cam = torch.relu(cam)

        H = xdict[f'x_{largest}'].shape[-2]
        W = xdict[f'x_{largest}'].shape[-1]
        cam = nn.functional.interpolate(cam, size=(H, W), mode='bilinear', align_cc
        cam = cam[0, 0].detach().cpu().numpy()

        h1.remove()
        h2.remove()

        cam = cam - cam.min()
        cam = cam / (cam.max() + 1e-9)
        return cam, largest

    print("Improved Grad-CAM helpers ready.")
```

```
Improved Grad-CAM helpers ready.
```

```
# ============================================================
# OPTION B — TAIL CELL 4
# Run improved Grad-CAM on the three finalists
# Run this next, then paste the three summary tables here.
# ============================================================

def evaluate_gradcam_groups_b(final_model, final_model_type, final_pred_df, tai
                              source_include_ma, source_include_volume,
                              scales_if_multi=None, max_eval=None):
    y_prob = final_pred_df['prob_up'].values
    low_thr = np.quantile(y_prob, tail_q)
    high_thr = np.quantile(y_prob, 1 - tail_q)
    keep = (y_prob <= low_thr) | (y_prob >= high_thr)

    df = final_pred_df.loc[keep].copy().reset_index(drop=True)
    if max_eval is not None:
        df = df.head(max_eval).copy()

    rows = []
    for _, r in tqdm(df.iterrows(), total=len(df), desc=f'Grad-CAM {final_model
        sid = int(r['sample_id'])
        sample_row = master_b_q.loc[master_b_q['sample_id'] == sid].iloc[0]

        if final_model_type == 'single_scale':
            scale = int(r['scale'])
            img = render_refstyle_window(
                cf.iloc[int(sample_row['anchor_idx']) - scale + 1: int(sample_r
                scale=scale,
                include_ma=source_include_ma,
                include_volume=source_include_volume,
            )
            x = to_tensor_b(img).unsqueeze(0).to(DEVICE)
            cam = gradcam_single_b(final_model, x)
            img_arr = np.array(img)

        else:
            scales_local = scales_if_multi if scales_if_multi is not None else
            xdict = {}
            imgs = {}
            for s in scales_local:
                img = render_refstyle_window(
                    cf.iloc[int(sample_row['anchor_idx']) - s + 1: int(sample_r
                    scale=s,
                    include_ma=source_include_ma,
                    include_volume=source_include_volume,
                )
                imgs[s] = img
                xdict[f'x_{s}'] = to_tensor_b(img).unsqueeze(0).to(DEVICE)
```

```
                cam, largest = gradcam_multi_b(final_model, xdict)
                img_arr = np.array(imgs[largest])

            masks = build_structured_masks_from_image(Image.fromarray(img_arr))
            total = cam.sum() + 1e-9

            dil_fg_heat = float((cam * masks['fg_dil']).sum() / total)
            bg_heat = float((cam * masks['bg_mask']).sum() / total)
            price_fg_heat = float((cam * masks['price_fg']).sum() / total)
            volume_fg_heat = float((cam * masks['volume_fg']).sum() / total)
            recent_heat = float((cam * masks['recent_mask']).sum() / total)

            focus_group = 'chart_focused' if dil_fg_heat >= bg_heat else 'backgrour

            pred = 1 if r['prob_up'] >= high_thr else 0
            correct = int(pred == int(r['label_up']))

            rows.append({
                'sample_id': sid,
                'prob_up': float(r['prob_up']),
                'label_up': int(r['label_up']),
                'pred_tail': int(pred),
                'correct_tail': correct,
                'forward_return': float(r['forward_return']),
                'foreground_heat_dilated': dil_fg_heat,
                'background_heat': bg_heat,
                'price_foreground_heat': price_fg_heat,
                'volume_foreground_heat': volume_fg_heat,
                'recent_right_edge_heat': recent_heat,
                'focus_group': focus_group,
                'contract': r['contract'],
                'anchor_ts': r['anchor_ts'],
            })

        detail = pd.DataFrame(rows)
        summary = detail.groupby('focus_group').agg(
            rows=('sample_id', 'count'),
            accuracy=('correct_tail', 'mean'),
            avg_prob=('prob_up', 'mean'),
            avg_fg_heat_dilated=('foreground_heat_dilated', 'mean'),
            avg_bg_heat=('background_heat', 'mean'),
            avg_price_fg_heat=('price_foreground_heat', 'mean'),
            avg_volume_fg_heat=('volume_foreground_heat', 'mean'),
            avg_recent_right_edge_heat=('recent_right_edge_heat', 'mean'),
            avg_forward_return=('forward_return', 'mean'),
        ).reset_index()

        return detail, summary

    # ---- helper to pick model object by run_name ----
```

```
run_lookup = {
    'single_I120_ma0_vol0': run_I120_ohcl,
    'multi_gated_20_60_120_ma0_vol0': run_multi_gated_ohcl,
    'multi_gated_20_60_120_ma1_vol1': run_multi_gated_full,
}

gradcam_optionB_outputs = {}

for _, row in finalists_df.iterrows():
    role = row['role']
    source = row['source']
    tail_q = float(row['tail_q_each_side'])

    out = run_lookup[source]
    final_model = out['model_obj']
    final_model_type = out['model_type']
    final_pred_df = out['test_pred']
    source_include_ma = bool(out['include_ma'])
    source_include_volume = bool(out['include_volume'])
    scales_if_multi = out.get('scales', None)

    print(f"\n===== Running Grad-CAM for {role} | {source} | q={tail_q} =====")
    detail_df, summary_df = evaluate_gradcam_groups_b(
        final_model=final_model,
        final_model_type=final_model_type,
        final_pred_df=final_pred_df,
        tail_q=tail_q,
        source_include_ma=source_include_ma,
        source_include_volume=source_include_volume,
        scales_if_multi=scales_if_multi,
        max_eval=None
    )

    detail_fp = RESULT_DIR / f'optionB_gradcam_detail_{role}.csv'
    summary_fp = RESULT_DIR / f'optionB_gradcam_summary_{role}.csv'
    detail_df.to_csv(detail_fp, index=False)
    summary_df.to_csv(summary_fp, index=False)

    gradcam_optionB_outputs[role] = {
        'source': source,
        'tail_q': tail_q,
        'detail': detail_df,
        'summary': summary_df,
        'detail_fp': str(detail_fp),
        'summary_fp': str(summary_fp),
    }

# show summaries
for role, obj in gradcam_optionB_outputs.items():
    print(f"\n===== SUMMARY: {role} | {obj['source']} | q={obj['tail_q']} =====
    display(obj['summary'])
```

```
===== Running Grad-CAM for strict_best_score | single_I120_ma0_vol0 | q=0.01 ===
Grad-                                                    54/54 [00:01<00:00, 31.88it/s]
CAM single_scale: 100%

===== Running Grad-CAM for best_multiscale | multi_gated_20_60_120_ma0_vol0 | q=
Grad-                                                    54/54 [00:01<00:00, 30.56it/s]
CAM multi_scale_gated: 100%

===== Running Grad-CAM for best_rich_fullimage | multi_gated_20_60_120_ma1_vol1
Grad-                                                    54/54 [00:01<00:00, 28.91it/s]
CAM multi_scale_gated: 100%

===== SUMMARY: strict_best_score | single_I120_ma0_vol0 | q=0.01 =====
```

	focus_group	rows	accuracy	avg_prob	avg_fg_heat_dilated	avg_bg_heat
0	background_focused	54	0.833333	0.48246	0.141516	0.858484

```
===== SUMMARY: best_multiscale | multi_gated_20_60_120_ma0_vol0 | q=0.01 =====
```

	focus_group	rows	accuracy	avg_prob	avg_fg_heat_dilated	avg_bg_heat
0	background_focused	54	0.796296	0.482469	0.22444	0.77556

```
# ==============================================================
# HOG — CELL 1
# HOG feature extraction dataset build
# Run this first, then paste the output here.
# ==============================================================

# If needed in Colab:
# !pip -q install scikit-image

from skimage.feature import hog

# ----------------------------
# Choose the dataset branch
# ----------------------------
# Start with I120 because it is now your strongest selective model.
HOG_SCALE = 120
HOG_INCLUDE_MA = False
HOG_INCLUDE_VOLUME = False

# Use the Option B master set if available, otherwise fall back
if 'master_b_q' in globals():
    hog_manifest = master_b_q.copy()
else:
```

```
    hog_manifest = master_q.copy()

print("Using manifest rows:", len(hog_manifest))
print("Using HOG scale:", HOG_SCALE)
print("Include MA:", HOG_INCLUDE_MA, "| Include volume:", HOG_INCLUDE_VOLUME)

# ----------------------------
# HOG extraction settings
# ----------------------------
HOG_PARAMS = {
    'orientations': 9,
    'pixels_per_cell': (8, 8),
    'cells_per_block': (2, 2),
    'block_norm': 'L2-Hys',
    'transform_sqrt': True,
    'feature_vector': True,
}

def render_image_for_sample(sample_row, scale=120, include_ma=False, include_vc
    a = int(sample_row['anchor_idx'])
    window = cf.iloc[a - scale + 1: a + 1].copy()
    img = render_refstyle_window(
        window,
        scale=scale,
        include_ma=include_ma,
        include_volume=include_volume
    )
    return np.array(img)

def compute_hog_features(img_arr, hog_params=HOG_PARAMS):
    # img_arr expected as grayscale uint8 array
    feats = hog(img_arr, **hog_params)
    return feats.astype(np.float32)

# ----------------------------
# Extract HOG features
# ----------------------------
hog_rows = []
hog_features = []

for _, r in tqdm(hog_manifest.iterrows(), total=len(hog_manifest), desc="Extrac
    img_arr = render_image_for_sample(
        r,
        scale=HOG_SCALE,
        include_ma=HOG_INCLUDE_MA,
        include_volume=HOG_INCLUDE_VOLUME
    )
    feats = compute_hog_features(img_arr)

    hog_rows.append({
        'sample_id': int(r['sample_id']),
```

```
            'split': r['split'],
            'label_up': int(r['label_up']),
            'forward_return': float(r['forward_return']),
            'contract': str(r['contract']),
            'anchor_ts': str(r['anchor_ts']),
        })
        hog_features.append(feats)

    hog_meta_df = pd.DataFrame(hog_rows)
    hog_X = np.vstack(hog_features)

    print("hog_meta_df shape:", hog_meta_df.shape)
    print("hog_X shape:", hog_X.shape)
    print("train rows:", (hog_meta_df['split'] == 'train').sum())
    print("val rows:", (hog_meta_df['split'] == 'val').sum())
    print("test rows:", (hog_meta_df['split'] == 'test').sum())
    print("positive rate train:", hog_meta_df.loc[hog_meta_df['split']=='train', ']

    # Save for reuse
    np.save(RESULT_DIR / f'hog_X_I{HOG_SCALE}_ma{int(HOG_INCLUDE_MA)}_vol{int(HOG_I
    hog_meta_df.to_csv(RESULT_DIR / f'hog_meta_I{HOG_SCALE}_ma{int(HOG_INCLUDE_MA)}

    print("Saved HOG arrays to results folder.")
```

```
Using manifest rows: 17013
Using HOG scale: 120
Include MA: False | Include volume: False
Extracting HOG features: 100%|| 17013/17013 [07:21<00:00, 43.48it/s]
hog_meta_df shape: (17013, 6)
hog_X shape: (17013, 23760)
train rows: 11745
val rows: 2581
test rows: 2687
positive rate train: 0.4808003405704555
Saved HOG arrays to results folder.
```

```
# ============================================================
# HOG — CELL 2
# HOG benchmark model: SVD + Logistic Regression
# Run this next, then paste the output here.
# ============================================================

from sklearn.decomposition import TruncatedSVD
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression

# ----------------------------
# Train / val / test split
# ----------------------------
mask_train = hog_meta_df['split'].values == 'train'
mask_val   = hog_meta_df['split'].values == 'val'
```

```
mask_test  = hog_meta_df['split'].values == 'test'

X_train = hog_X[mask_train]
X_val   = hog_X[mask_val]
X_test  = hog_X[mask_test]

y_train = hog_meta_df.loc[mask_train, 'label_up'].values.astype(int)
y_val   = hog_meta_df.loc[mask_val, 'label_up'].values.astype(int)
y_test  = hog_meta_df.loc[mask_test, 'label_up'].values.astype(int)

print("X_train:", X_train.shape, "| X_val:", X_val.shape, "| X_test:", X_test.s

# ----------------------------
# Scale + reduce dimensionality
# ----------------------------
# HOG vectors are large; this makes the benchmark more stable and faster.
hog_scaler = StandardScaler(with_mean=False)
X_train_s = hog_scaler.fit_transform(X_train)
X_val_s   = hog_scaler.transform(X_val)
X_test_s  = hog_scaler.transform(X_test)

HOG_SVD_DIM = 256
hog_svd = TruncatedSVD(n_components=HOG_SVD_DIM, random_state=SEED)

X_train_z = hog_svd.fit_transform(X_train_s)
X_val_z   = hog_svd.transform(X_val_s)
X_test_z  = hog_svd.transform(X_test_s)

print("Reduced HOG shapes:", X_train_z.shape, X_val_z.shape, X_test_z.shape)
print("Explained variance ratio sum:", float(hog_svd.explained_variance_ratio_.

# ----------------------------
# Logistic regression on reduced HOG features
# ----------------------------
hog_logit = LogisticRegression(
    max_iter=2000,
    class_weight='balanced',
    solver='lbfgs',
    random_state=SEED
)
hog_logit.fit(X_train_z, y_train)

val_prob = hog_logit.predict_proba(X_val_z)[:, 1]
test_prob = hog_logit.predict_proba(X_test_z)[:, 1]

# threshold chosen on validation only
hog_best_thr, hog_best_val_score = choose_threshold(
    y_val, val_prob, metric=THRESHOLD_METRIC
)

hog_val_metrics = metric_bundle(y_val, val_prob, threshold=hog_best_thr)
```

```
hog_test_metrics = metric_bundle(y_test, test_prob, threshold=hog_best_thr)

print("\n=== HOG BENCHMARK: VALIDATION METRICS ===")
for k, v in hog_val_metrics.items():
    print(f"{k}: {v}")

print("\n=== HOG BENCHMARK: TEST METRICS ===")
for k, v in hog_test_metrics.items():
    print(f"{k}: {v}")

# ----------------------------
# Tail-rule evaluation
# ----------------------------
hog_tail_rows = []
for q in [0.01, 0.015, 0.025, 0.05, 0.10]:
    out = evaluate_tail_rule(y_test, test_prob, top_q=q, bottom_q=q)
    out['tail_q_each_side'] = q
    out['source'] = f'HOG_I{HOG_SCALE}_ma{int(HOG_INCLUDE_MA)}_vol{int(HOG_INCL
    hog_tail_rows.append(out)

hog_tail_df = pd.DataFrame(hog_tail_rows).sort_values(
    ['balanced_accuracy', 'mcc', 'coverage'],
    ascending=False
).reset_index(drop=True)

print("\n=== HOG TAIL-RULE RESULTS ===")
display(hog_tail_df)

# ----------------------------
# Save
# ----------------------------
hog_pred_df = hog_meta_df.loc[mask_test, ['sample_id', 'contract', 'forward_ret
hog_pred_df['prob_up'] = test_prob
hog_pred_df['run_name'] = f'HOG_I{HOG_SCALE}_ma{int(HOG_INCLUDE_MA)}_vol{int(HO

hog_tail_df.to_csv(RESULT_DIR / f'hog_tail_results_I{HOG_SCALE}_ma{int(HOG_INCL
hog_pred_df.to_csv(RESULT_DIR / f'hog_test_pred_I{HOG_SCALE}_ma{int(HOG_INCLUDE

print("\nSaved HOG benchmark outputs to results folder.")
print("Chosen threshold:", hog_best_thr)
print("Best validation score on", THRESHOLD_METRIC, "=", hog_best_val_score)
```

```
X_train: (11745, 23760) | X_val: (2581, 23760) | X_test: (2687, 23760)
Reduced HOG shapes: (11745, 256) (2581, 256) (2687, 256)
Explained variance ratio sum: 0.6605235934257507
```

```
# ============================================================
# HAAR — CELL 1
# Build HAAR-like / rectangle-feature dataset
# Run this first, then paste the output here.
# ============================================================

# We start from the strongest current scale for fair comparison
HAAR_SCALE = 120
HAAR_INCLUDE_MA = False
HAAR_INCLUDE_VOLUME = False

# Use Option B manifest if available
if 'master_b_q' in globals():
    haar_manifest = master_b_q.copy()
else:
    haar_manifest = master_q.copy()

print("Using manifest rows:", len(haar_manifest))
print("Using HAAR scale:", HAAR_SCALE)
print("Include MA:", HAAR_INCLUDE_MA, "| Include volume:", HAAR_INCLUDE_VOLUME)

def render_image_for_haar(sample_row, scale=120, include_ma=False, include_volu
    a = int(sample_row['anchor_idx'])
    window = cf.iloc[a - scale + 1: a + 1].copy()
    img = render_refstyle_window(
        window,
        scale=scale,
        include_ma=include_ma,
        include_volume=include_volume
    )
    return np.array(img).astype(np.float32)

def integral_image(arr):
    # padded integral image
    ii = np.zeros((arr.shape[0] + 1, arr.shape[1] + 1), dtype=np.float64)
    ii[1:, 1:] = arr.cumsum(axis=0).cumsum(axis=1)
    return ii

def rect_sum(ii, r0, c0, r1, c1):
    # rectangle [r0:r1, c0:c1)
    return ii[r1, c1] - ii[r0, c1] - ii[r1, c0] + ii[r0, c0]

def compute_haar_like_features(img_arr, price_panel_ratio=PRICE_PANEL_RATIO):
    """
    Simple thesis-friendly HAAR-like rectangle features.
    They measure contrasts between structured chart regions.
    """
```

```
    H, W = img_arr.shape
    price_h = int(round(H * price_panel_ratio))
    vol_h = H - price_h

    ii = integral_image(img_arr)

    feats = {}

    # ----------------------------
    # Whole-image / panel contrasts
    # ----------------------------
    left_sum  = rect_sum(ii, 0, 0, H, W // 2)
    right_sum = rect_sum(ii, 0, W // 2, H, W)

    top_sum   = rect_sum(ii, 0, 0, price_h // 2, W)
    mid_sum   = rect_sum(ii, price_h // 2, 0, price_h, W)

    price_sum = rect_sum(ii, 0, 0, price_h, W)
    vol_sum   = rect_sum(ii, price_h, 0, H, W)

    feats['left_minus_right'] = float(left_sum - right_sum)
    feats['right_minus_left'] = float(right_sum - left_sum)
    feats['top_minus_mid_price'] = float(top_sum - mid_sum)
    feats['mid_minus_top_price'] = float(mid_sum - top_sum)
    feats['price_minus_volume'] = float(price_sum - vol_sum)
    feats['volume_minus_price'] = float(vol_sum - price_sum)

    # ----------------------------
    # Recent-right-edge vs earlier-left
    # ----------------------------
    recent_w = max(15, W // 8)
    recent_sum = rect_sum(ii, 0, W - recent_w, price_h, W)
    early_sum  = rect_sum(ii, 0, 0, price_h, recent_w)

    feats['recent_price_minus_early_price'] = float(recent_sum - early_sum)
    feats['early_price_minus_recent_price'] = float(early_sum - recent_sum)

    # ----------------------------
    # Horizontal stripe contrasts in price panel
    # ----------------------------
    stripe_h = max(1, price_h // 4)
    s1 = rect_sum(ii, 0, 0, stripe_h, W)
    s2 = rect_sum(ii, stripe_h, 0, min(2 * stripe_h, price_h), W)
    s3 = rect_sum(ii, min(2 * stripe_h, price_h), 0, min(3 * stripe_h, price_h)
    s4 = rect_sum(ii, min(3 * stripe_h, price_h), 0, price_h, W)

    feats['stripe1_minus_stripe4'] = float(s1 - s4)
    feats['stripe2_minus_stripe3'] = float(s2 - s3)
    feats['upper_half_minus_lower_half_price'] = float((s1 + s2) - (s3 + s4))

    # ----------------------------
```

```
        # Vertical stripe contrasts in price panel
        # ----------------------------
        stripe_w = max(1, W // 4)
        v1 = rect_sum(ii, 0, 0, price_h, stripe_w)
        v2 = rect_sum(ii, 0, stripe_w, price_h, min(2 * stripe_w, W))
        v3 = rect_sum(ii, 0, min(2 * stripe_w, W), price_h, min(3 * stripe_w, W))
        v4 = rect_sum(ii, 0, min(3 * stripe_w, W), price_h, W)

        feats['vstripe1_minus_vstripe4'] = float(v1 - v4)
        feats['vstripe2_minus_vstripe3'] = float(v2 - v3)
        feats['left_half_minus_right_half_price'] = float((v1 + v2) - (v3 + v4))

        # ----------------------------
        # Two-rectangle edge features near the right side
        # ----------------------------
        right_block_w = max(10, W // 6)
        upper_right = rect_sum(ii, 0, W - right_block_w, price_h // 2, W)
        lower_right = rect_sum(ii, price_h // 2, W - right_block_w, price_h, W)

        feats['upper_right_minus_lower_right'] = float(upper_right - lower_right)
        feats['lower_right_minus_upper_right'] = float(lower_right - upper_right)

        # ----------------------------
        # Volume structure if volume exists
        # ----------------------------
        if vol_h > 0:
            vol_left  = rect_sum(ii, price_h, 0, H, W // 2)
            vol_right = rect_sum(ii, price_h, W // 2, H, W)
            feats['vol_left_minus_vol_right'] = float(vol_left - vol_right)
            feats['vol_right_minus_vol_left'] = float(vol_right - vol_left)
        else:
            feats['vol_left_minus_vol_right'] = 0.0
            feats['vol_right_minus_vol_left'] = 0.0

        return feats

    # ----------------------------
    # Extract HAAR-like features
    # ----------------------------
    haar_rows = []
    haar_feat_rows = []

    for _, r in tqdm(haar_manifest.iterrows(), total=len(haar_manifest), desc="Extr
        img_arr = render_image_for_haar(
            r,
            scale=HAAR_SCALE,
            include_ma=HAAR_INCLUDE_MA,
            include_volume=HAAR_INCLUDE_VOLUME
        )
        feats = compute_haar_like_features(img_arr)
```

```
        meta = {
            'sample_id': int(r['sample_id']),
            'split': r['split'],
            'label_up': int(r['label_up']),
            'forward_return': float(r['forward_return']),
            'contract': str(r['contract']),
            'anchor_ts': str(r['anchor_ts']),
        }

        haar_rows.append(meta)
        haar_feat_rows.append(feats)

    haar_meta_df = pd.DataFrame(haar_rows)
    haar_feat_df = pd.DataFrame(haar_feat_rows)
    haar_df = pd.concat([haar_meta_df, haar_feat_df], axis=1)

    print("haar_df shape:", haar_df.shape)
    print("Feature columns:", len(haar_feat_df.columns))
    print("train rows:", (haar_df['split'] == 'train').sum())
    print("val rows:", (haar_df['split'] == 'val').sum())
    print("test rows:", (haar_df['split'] == 'test').sum())

    display(haar_df.head())

    haar_df.to_csv(
        RESULT_DIR / f'haar_like_features_I{HAAR_SCALE}_ma{int(HAAR_INCLUDE_MA)}_vc
        index=False
    )

    print("Saved HAAR-like feature table to results folder.")
```

```
Using manifest rows: 17013
Using HAAR scale: 120
Include MA: False | Include volume: False
```

Extracting HAAR-like features: 100% 17013/17013 [02:23<00:00, 95.50it/s]

```
haar_df shape: (17013, 24)
Feature columns: 18
train rows: 11745
val rows: 2581
test rows: 2687
```

	sample_id	split	label_up	forward_return	contract	anchor_ts	left_minus_r
0	0	train	0	-0.002536	SiH4	2024-01-08 08:45:00	186
1	1	train	0	-0.001574	SiH4	2024-01-08 09:00:00	153
2	3	train	1	0.002057	SiH4	2024-01-08 09:30:00	109
3	4	train	1	0.006292	SiH4	2024-01-08 09:45:00	163
4	5	train	1	0.004579	SiH4	2024-01-08 10:00:00	96

5 rows × 24 columns

```
Saved HAAR-like feature table to results folder
```

```
# ============================================================
# HAAR — CELL 2
# HAAR-like benchmark model: Logistic Regression
# Run this next, then paste the output here.
# ============================================================

from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression

# ----------------------------
# Prepare train / val / test
# ----------------------------
meta_cols = ['sample_id', 'split', 'label_up', 'forward_return', 'contract', 'a
haar_feature_cols = [c for c in haar_df.columns if c not in meta_cols]

train_df_haar = haar_df[haar_df['split'] == 'train'].copy()
val_df_haar   = haar_df[haar_df['split'] == 'val'].copy()
```

```
test_df_haar  = haar_df[haar_df['split'] == 'test'].copy()

X_train = train_df_haar[haar_feature_cols].values.astype(np.float32)
X_val   = val_df_haar[haar_feature_cols].values.astype(np.float32)
X_test  = test_df_haar[haar_feature_cols].values.astype(np.float32)

y_train = train_df_haar['label_up'].values.astype(int)
y_val   = val_df_haar['label_up'].values.astype(int)
y_test  = test_df_haar['label_up'].values.astype(int)

print("X_train:", X_train.shape, "| X_val:", X_val.shape, "| X_test:", X_test.s

# ----------------------------
# Scale features
# ----------------------------
haar_scaler = StandardScaler()
X_train_s = haar_scaler.fit_transform(X_train)
X_val_s   = haar_scaler.transform(X_val)
X_test_s  = haar_scaler.transform(X_test)

# ----------------------------
# Logistic regression
# ----------------------------
haar_logit = LogisticRegression(
    max_iter=2000,
    class_weight='balanced',
    solver='lbfgs',
    random_state=SEED
)
haar_logit.fit(X_train_s, y_train)

val_prob = haar_logit.predict_proba(X_val_s)[:, 1]
test_prob = haar_logit.predict_proba(X_test_s)[:, 1]

# choose threshold on validation only
haar_best_thr, haar_best_val_score = choose_threshold(
    y_val, val_prob, metric=THRESHOLD_METRIC
)

haar_val_metrics = metric_bundle(y_val, val_prob, threshold=haar_best_thr)
haar_test_metrics = metric_bundle(y_test, test_prob, threshold=haar_best_thr)

print("\n=== HAAR-LIKE BENCHMARK: VALIDATION METRICS ===")
for k, v in haar_val_metrics.items():
    print(f"{k}: {v}")

print("\n=== HAAR-LIKE BENCHMARK: TEST METRICS ===")
for k, v in haar_test_metrics.items():
    print(f"{k}: {v}")

# ----------------------------
```

```
# Tail-rule evaluation
# ----------------------------
haar_tail_rows = []
for q in [0.01, 0.015, 0.025, 0.05, 0.10]:
    out = evaluate_tail_rule(y_test, test_prob, top_q=q, bottom_q=q)
    out['tail_q_each_side'] = q
    out['source'] = f'HAAR_I{HAAR_SCALE}_ma{int(HAAR_INCLUDE_MA)}_vol{int(HAAR_
    haar_tail_rows.append(out)

haar_tail_df = pd.DataFrame(haar_tail_rows).sort_values(
    ['balanced_accuracy', 'mcc', 'coverage'],
    ascending=False
).reset_index(drop=True)

print("\n=== HAAR-LIKE TAIL-RULE RESULTS ===")
display(haar_tail_df)

# ----------------------------
# Save
# ----------------------------
haar_pred_df = test_df_haar[['sample_id', 'contract', 'forward_return', 'anchor
haar_pred_df['prob_up'] = test_prob
haar_pred_df['run_name'] = f'HAAR_I{HAAR_SCALE}_ma{int(HAAR_INCLUDE_MA)}_vol{ir

haar_tail_df.to_csv(RESULT_DIR / f'haar_tail_results_I{HAAR_SCALE}_ma{int(HAAR_
haar_pred_df.to_csv(RESULT_DIR / f'haar_test_pred_I{HAAR_SCALE}_ma{int(HAAR_INC

print("\nSaved HAAR-like benchmark outputs to results folder.")
print("Chosen threshold:", haar_best_thr)
print("Best validation score on", THRESHOLD_METRIC, "=", haar_best_val_score)
```

```
X_train: (11745, 18) | X_val: (2581, 18) | X_test: (2687, 18)
```

```
# ============================================================
# XAI FIX — CELL 1
# Tight crop + explicit foreground mask channel
# ============================================================

from PIL import Image, ImageFilter
from torchvision import transforms

to_tensor_fix = transforms.Compose([transforms.ToTensor()])

def render_refstyle_window_cropped_with_mask(
    window_df,
    scale,
    include_ma=False,
    include_volume=False,
    crop_pad_px=6,
):
    """
    1) render the normal thesis image
    2) find actual visible foreground
    3) crop tightly around it with small padding
    4) resize back to the original canvas
    5) return both grayscale image and binary/dilated mask
    """
    base_img = render_refstyle_window(
        window_df,
        scale=scale,
        include_ma=include_ma,
        include_volume=include_volume
    )
    arr = np.array(base_img).astype(np.uint8)
    H, W = arr.shape

    fg_exact = (arr > 0).astype(np.uint8) * 255

    ys, xs = np.where(fg_exact > 0)
    if len(ys) == 0 or len(xs) == 0:
        # fallback: no visible structure
        cropped_img = Image.fromarray(arr)
        cropped_mask = Image.fromarray(fg_exact)
    else:
        y0 = max(0, int(ys.min()) - crop_pad_px)
        y1 = min(H, int(ys.max()) + crop_pad_px + 1)
        x0 = max(0, int(xs.min()) - crop_pad_px)
        x1 = min(W, int(xs.max()) + crop_pad_px + 1)

        cropped_img = Image.fromarray(arr[y0:y1, x0:x1]).resize((W, H), Image.E
        cropped_mask = Image.fromarray(fg_exact[y0:y1, x0:x1]).resize((W, H), I
```

```
    # dilate the mask slightly so near-candle heat is not unfairly called backg
    cropped_mask = cropped_mask.filter(ImageFilter.MaxFilter(size=7))

    return cropped_img, cropped_mask


class CropMaskChartDataset(Dataset):
    def __init__(self, manifest_df, cf_df, scale, include_ma=False, include_vol
        self.df = manifest_df.reset_index(drop=True)
        self.cf = cf_df
        self.scale = scale
        self.include_ma = include_ma
        self.include_volume = include_volume

    def __len__(self):
        return len(self.df)

    def __getitem__(self, idx):
        r = self.df.iloc[idx]
        sid = int(r['sample_id'])
        a = int(r['anchor_idx'])
        window = self.cf.iloc[a - self.scale + 1: a + 1].copy()

        img_pil, mask_pil = render_refstyle_window_cropped_with_mask(
            window_df=window,
            scale=self.scale,
            include_ma=self.include_ma,
            include_volume=self.include_volume,
            crop_pad_px=6
        )

        x_img = to_tensor_fix(img_pil)      # [1,H,W]
        x_msk = to_tensor_fix(mask_pil)     # [1,H,W]
        x = torch.cat([x_img, x_msk], dim=0)  # [2,H,W]

        y = torch.tensor(int(r['label_up']), dtype=torch.long)

        return {
            'x': x,
            'y': y,
            'sample_id': torch.tensor(sid, dtype=torch.long),
        }


# quick smoke test
tmp_fix_ds = CropMaskChartDataset(master_b_q.head(2), cf, scale=120, include_ma
tmp_fix = tmp_fix_ds[0]

print("Crop-mask dataset ready.")
```

```
print("x shape:", tmp_fix['x'].shape)   # should be [2, H, W]
print("y:", tmp_fix['y'].item())
```

```
Crop-mask dataset ready.
x shape: torch.Size([2, 128, 360])
y: 0
```

```
# ============================================================
# XAI FIX — CELL 2
# 2-channel CNN for cropped image + foreground mask
# Start with the strongest current branch: I120 OHLC-only
# ============================================================

class CNNFeatureExtractor2Ch(nn.Module):
    def __init__(self, in_channels=2, base=32, depth=5):
        super().__init__()
        channels = []
        c = base
        for _ in range(depth):
            channels.append(c)
            c = min(c * 2, 256)

        blocks = []
        prev = in_channels
        for c in channels:
            blocks.extend([
                nn.Conv2d(prev, c, kernel_size=3, padding=1, bias=False),
                nn.BatchNorm2d(c),
                nn.LeakyReLU(0.01, inplace=True),
                nn.MaxPool2d(2, 2),
            ])
            prev = c

        self.backbone = nn.Sequential(*blocks)
        self.pool = nn.AdaptiveAvgPool2d((4, 4))
        self.out_dim = prev * 4 * 4

    def forward(self, x):
        z = self.backbone(x)
        z = self.pool(z)
        return torch.flatten(z, 1)


class CropMaskCNN(nn.Module):
    def __init__(self, dropout=0.5):
        super().__init__()
        self.fe = CNNFeatureExtractor2Ch(in_channels=2, base=32, depth=5)
        self.head = nn.Sequential(
            nn.Dropout(dropout),
            nn.Linear(self.fe.out_dim, 2),
        )
```

```
    def forward(self, x):
        return self.head(self.fe(x))


def run_epoch_cropmask(model, loader, optimizer=None, class_weights=None):
    train_mode = optimizer is not None
    model.train(train_mode)
    criterion = nn.CrossEntropyLoss(weight=class_weights)

    losses, probs, ys, ids = [], [], [], []

    for batch in tqdm(loader, leave=False):
        x = batch['x'].to(DEVICE, non_blocking=True)
        y = batch['y'].to(DEVICE, non_blocking=True)

        with torch.set_grad_enabled(train_mode):
            logits = model(x)
            loss = criterion(logits, y)
            if train_mode:
                optimizer.zero_grad(set_to_none=True)
                loss.backward()
                optimizer.step()

        p = torch.softmax(logits, dim=1)[:, 1].detach().cpu().numpy()
        losses.append(float(loss.item()))
        probs.extend(p.tolist())
        ys.extend(batch['y'].cpu().numpy().tolist())
        ids.extend(batch['sample_id'].cpu().numpy().tolist())

    return {
        'loss': float(np.mean(losses)),
        'y_true': np.array(ys),
        'y_prob': np.array(probs),
        'sample_id': np.array(ids),
    }


def train_cropmask_model(
    manifest_df,
    scale=120,
    include_ma=False,
    include_volume=False,
    lr=1e-4,
    weight_decay=1e-4,
    max_epochs=20,
    patience=5,
    run_name='single_I120_cropmask_ma0_vol0'
):
    train_df = manifest_df[manifest_df['split'] == 'train'].copy()
    val_df   = manifest_df[manifest_df['split'] == 'val'].copy()
```

```
    test_df  = manifest_df[manifest_df['split'] == 'test'].copy()

    ds_train = CropMaskChartDataset(train_df, cf, scale=scale, include_ma=inclu
    ds_val   = CropMaskChartDataset(val_df, cf, scale=scale, include_ma=include
    ds_test  = CropMaskChartDataset(test_df, cf, scale=scale, include_ma=incluc

    dl_train = DataLoader(ds_train, batch_size=BATCH_SIZE, shuffle=True,  num_w
    dl_val   = DataLoader(ds_val,   batch_size=BATCH_SIZE, shuffle=False, num_w
    dl_test  = DataLoader(ds_test,  batch_size=BATCH_SIZE, shuffle=False, num_w

    model = CropMaskCNN().to(DEVICE)
    optimizer = torch.optim.AdamW(model.parameters(), lr=lr, weight_decay=weigh

    pos = train_df['label_up'].mean()
    w0 = 1.0 / max(1.0 - pos, 1e-6)
    w1 = 1.0 / max(pos, 1e-6)
    class_weights = torch.tensor([w0, w1], dtype=torch.float32, device=DEVICE)

    best_auc = -np.inf
    best_state = None
    best_epoch = -1
    bad = 0

    for epoch in range(1, max_epochs + 1):
        tr = run_epoch_cropmask(model, dl_train, optimizer=optimizer, class_wei
        va = run_epoch_cropmask(model, dl_val, optimizer=None, class_weights=cl

        tr_m = metric_bundle(tr['y_true'], tr['y_prob'], threshold=0.5)
        va_m = metric_bundle(va['y_true'], va['y_prob'], threshold=0.5)

        print(f"[{run_name}] epoch {epoch:02d} | train_auc={tr_m['roc_auc']:.4f

        if va_m['roc_auc'] > best_auc:
            best_auc = va_m['roc_auc']
            best_state = {k: v.detach().cpu() for k, v in model.state_dict().it
            best_epoch = epoch
            bad = 0
        else:
            bad += 1
            if bad >= patience:
                print(f"[{run_name}] early stopping at epoch {epoch}")
                break

    model.load_state_dict(best_state)

    va = run_epoch_cropmask(model, dl_val, optimizer=None, class_weights=class_
    te = run_epoch_cropmask(model, dl_test, optimizer=None, class_weights=class

    best_thr, _ = choose_threshold(va['y_true'], va['y_prob'], metric=THRESHOLD
    val_metrics  = metric_bundle(va['y_true'], va['y_prob'], threshold=best_thr
    test_metrics = metric_bundle(te['y_true'], te['y_prob'], threshold=best_thr
```

```
    out_pred = test_df[['sample_id', 'anchor_ts', 'contract', 'forward_return',
    out_pred['prob_up'] = te['y_prob']
    out_pred['run_name'] = run_name
    out_pred['scale'] = scale
    out_pred['include_ma'] = include_ma
    out_pred['include_volume'] = include_volume

    ckpt_fp = MODEL_DIR / f'{run_name}.pt'
    torch.save(best_state, ckpt_fp)

    return {
        'run_name': run_name,
        'model_type': 'single_scale_cropmask',
        'scale': scale,
        'include_ma': include_ma,
        'include_volume': include_volume,
        'best_epoch': best_epoch,
        'best_val_auc': best_auc,
        'threshold': best_thr,
        'val_metrics': val_metrics,
        'test_metrics': test_metrics,
        'test_pred': out_pred,
        'ckpt_fp': str(ckpt_fp),
        'model_obj': model,
    }

print("Crop-mask model/trainer ready.")
```

```
Crop-mask model/trainer ready.
```

```
# ============================================================
# XAI FIX — CELL 3
# Train cropped + mask-guided I120 OHLC-only model
# ============================================================

run_cropmask_I120 = train_cropmask_model(
    manifest_df=master_b_q,
    scale=120,
    include_ma=False,
    include_volume=False,
    lr=1e-4,
    weight_decay=WEIGHT_DECAY,
    max_epochs=MAX_EPOCHS,
    patience=PATIENCE,
    run_name='single_I120_cropmask_ma0_vol0'
)

print("\n=== FULL-SAMPLE TEST METRICS: single_I120_cropmask_ma0_vol0 ===")
for k, v in run_cropmask_I120['test_metrics'].items():
    print(f"{k}: {v}")
```

```
cropmask_pred_df = run_cropmask_I120['test_pred'].copy()

cropmask_tail_rows = []
for q in [0.01, 0.015, 0.025, 0.05, 0.10]:
    out = evaluate_tail_rule(
        cropmask_pred_df['label_up'].values.astype(int),
        cropmask_pred_df['prob_up'].values.astype(float),
        top_q=q, bottom_q=q
    )
    out['tail_q_each_side'] = q
    out['source'] = 'single_I120_cropmask_ma0_vol0'
    cropmask_tail_rows.append(out)

cropmask_tail_df = pd.DataFrame(cropmask_tail_rows).sort_values(
    ['balanced_accuracy', 'mcc', 'coverage'],
    ascending=False
).reset_index(drop=True)

print("\n=== TAIL-RULE RESULTS: single_I120_cropmask_ma0_vol0 ===")
display(cropmask_tail_df)
```

```
[single_I120_cropmask_ma0_vol0] epoch 01 | train_auc=0.5629 | val_auc=0.5591 |
```

```
[single_I120_cropmask_ma0_vol0] epoch 02 | train_auc=0.6383 | val_auc=0.5200 |
```